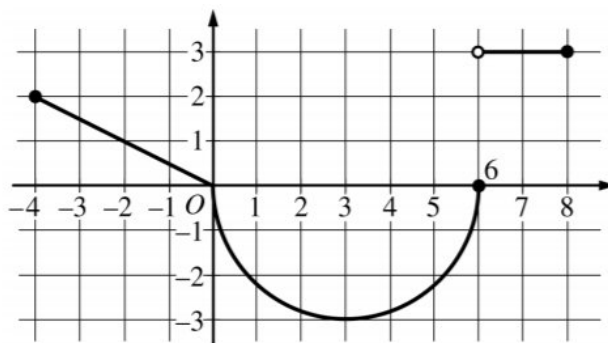


Unit 1 AP Topics 5-8

1. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of g

The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above. Let f be the function defined by $f(x) = 3x + \int_0^x g(t) dt$.

- (a) Find $f(7)$ and $f'(7)$.
- (b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.
- (c) For each of $\lim_{x \rightarrow 0^-} g'(x)$ and $\lim_{x \rightarrow 0^+} g'(x)$, find the value or state that it does not exist.
- (d) Find $\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1}$.

Part A

1 point is earned for: $f(7)$ 1 point is earned for: $f'(7)$

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $f(7)$ 1 point is earned for: $f'(7)$

Unit 1 AP Topics 5-8

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) \, dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$

Part B

2 points are earned for: answer with justification

On the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.

Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: answer with justification

On the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.

Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.

Part C

1 point is earned for: left-hand limit

1 point is earned for: right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} g'(x) \text{ does not exist.}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: left-hand limit

1 point is earned for: right-hand limit

Unit 1 AP Topics 5-8

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} g'(x) \text{ does not exist.}$$

Part D

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

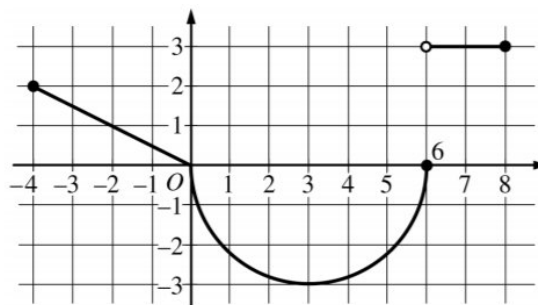
Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

Unit 1 AP Topics 5-8

2. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Graph of g

The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above. Let f be the function defined by $f(x) = 3x + \int_0^x g(t) dt$.

(a) Find $f(7)$ and $f'(7)$.

(b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.

(c) For each of $\lim_{x \rightarrow 0^-} g'(x)$ and $\lim_{x \rightarrow 0^+} g'(x)$, find the value or state that it does not exist.

(d) Find $\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1}$.

Part A

1 point is earned for: $f(7)$

1 point is earned for: $f'(7)$

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $f(7)$

1 point is earned for: $f'(7)$

$$f(7) = 3 \cdot 7 + \int_0^7 g(t) dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$$

$$f'(7) = 3 + g(7) = 3 + 3 = 6$$

Unit 1 AP Topics 5-8

Part B

2 points are earned for: answer with justification

On the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.

Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for: answer with justification

On the interval $-4 \leq x \leq 3$, $f'(x) = 3 + g(x)$.

Because $f'(x) \geq 0$ for $-4 \leq x \leq 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.

Part C

1 point is earned for: left-hand limit

1 point is earned for: right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} g'(x) \text{ does not exist.}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: left-hand limit

1 point is earned for: right-hand limit

$$\lim_{x \rightarrow 0^-} g'(x) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} g'(x) \text{ does not exist.}$$

Unit 1 AP Topics 5-8

Part D

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: limits equal 0

1 point is earned for: applies L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x \rightarrow -2} (f(x) + 7) = -6 + \int_0^{-2} g(t) dt + 7 = 0$$

$$\lim_{x \rightarrow -2} (e^{3x+6} - 1) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow -2} \frac{f(x)+7}{e^{3x+6}-1} = \lim_{x \rightarrow -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3+g(-2)}{3} = \frac{3+1}{3} = \frac{4}{3}.$$

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

Unit 1 AP Topics 5-8

3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Functions f , g , and h are twice-differentiable functions with $g(2) = h(2) = 4$. The line $y = 4 + \frac{2}{3}(x - 2)$ is tangent to both the graph of g at $x = 2$ and the graph of h at $x = 2$.

- Find $h'(2)$.
- Let a be the function given by $a(x) = 3x^3h(x)$. Write an expression for $a'(x)$. Find $a'(2)$.
- The function h satisfies $h(x) = \frac{x^2 - 4}{1 - (f(x))^3}$ for $x \neq 2$. It is known that $\lim_{x \rightarrow 2} h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim_{x \rightarrow 2} h(x)$ to find $f(2)$ and $f'(2)$. Show the work that leads to your answers.
- It is known that $g(x) \leq h(x)$ for $1 < x < 3$. Let k be a function satisfying $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$. Is k continuous at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
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The student response accurately includes the criteria below.

☐ · answer

Solution:

$$h'(2) = \frac{2}{3}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 5-8

The first point is earned for a derivative in the form of the product rule, namely $a'(x) = p(x) \cdot h(x) + q(x) \cdot h'(x)$, where $p(x)$ and $q(x)$ are both polynomials.

Simplification is not required to earn the third point. Function values must be substituted into the expression.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ form of product rule
- ☐ $a'(x)$
- ☐ $a'(2)$

Solution:

$$a'(x) = 9x^2h(x) + 3x^3h'(x)$$

$$a'(2) = 9 \cdot 2^2h(2) + 3 \cdot 2^3h'(2) = 36 \cdot 4 + 24 \cdot \frac{2}{3} = 160$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for showing $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$. This work may appear anywhere in the response.

The second point is earned for calculating $f(2) = 0$ in the context of setting the denominator $1 - (f(x))^3$ equal to 0.

The third point is earned for evidence of applying L'Hospital's Rule, which includes a new limit of a quotient where differentiation was attempted in both the numerator and denominator. Note that incorrect notation such as $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = \frac{0}{0}$ makes a response not eligible to earn the third point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$
- ☐ $f(2)$

Unit 1 AP Topics 5-8

- ☐ L'Hospital's Rule
- ☐ $f'(2)$

Solution:

Because h is differentiable, h is continuous, so $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Also, $\lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3}$, so $\lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3} = 4$.

Because $\lim_{x \rightarrow 2} (x^2 - 4) = 0$, we must also have $\lim_{x \rightarrow 2} (1 - (f(x))^3) = 0$.

Thus $\lim_{x \rightarrow 2} f(x) = 1$.

Because f is differentiable, f is continuous, so $f(2) = \lim_{x \rightarrow 2} f(x) = 1$.

Also, because f is twice differentiable, f' is continuous, so $\lim_{x \rightarrow 2} f'(x) = f'(2)$ exists.

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{1 - (f(x))^3} = \lim_{x \rightarrow 2} \frac{2x}{-3(f(x))^2 f'(x)} = \frac{4}{-3(1)^2 \cdot f'(2)} = 4.$$

Thus $f'(2) = -\frac{1}{3}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point does not require naming the squeeze theorem. The justification requires (1) stating g and h are continuous, (2) stating $\lim_{x \rightarrow 2} g(x) = 4$ and $\lim_{x \rightarrow 2} h(x) = 4$ and concluding that $\lim_{x \rightarrow 2} k(x) = 4$, and (3) stating $g(2) = 4 = h(2)$ and concluding $k(2) = 4$.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ continuous with justification

Solution:

Because g and h are differentiable, g and h are continuous, so $\lim_{x \rightarrow 2} g(x) = g(2) = 4$ and $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Unit 1 AP Topics 5-8

Because $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$, it follows from the squeeze theorem that $\lim_{x \rightarrow 2} k(x) = 4$.

Also, $4 = g(2) \leq k(2) \leq h(2) = 4$, so $k(2) = 4$.

Thus k is continuous at $x = 2$.

4. Which of the following limits are equal to -1 ?

I. $\lim_{x \rightarrow 0^-} \frac{|x|}{x}$

II. $\lim_{x \rightarrow 3} \frac{x^2 - 7x + 12}{3 - x}$

III. $\lim_{x \rightarrow \infty} \frac{1 - x}{1 + x}$

(A) I only

(B) I and III only

(C) II and III only

(D) I, II, and III only



Unit 1 AP Topics 5-8

5. Let f be the function that is given by $f(x) = ax + b/x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y -axis.

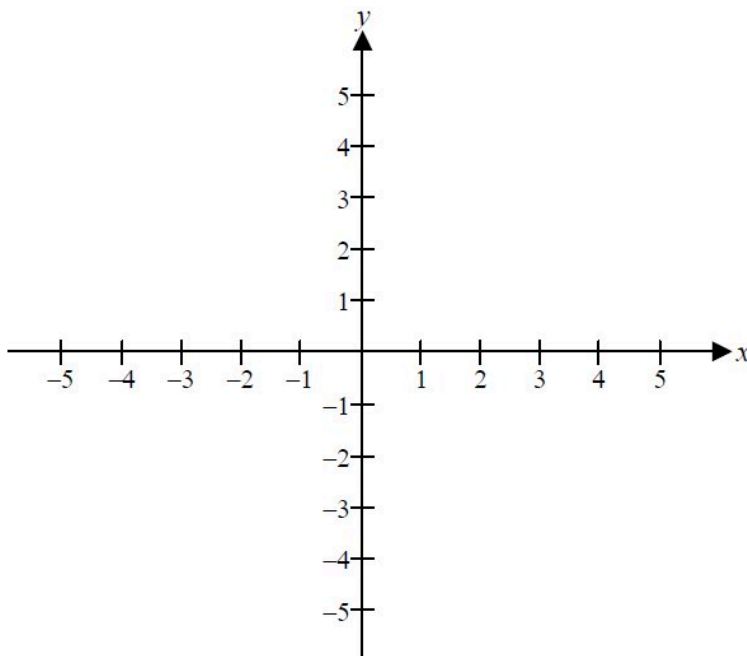
(ii) $\lim_{x \rightarrow 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

(a) Determine the values of a , b , and c .

(b) Write an equation for each vertical and each horizontal asymptote of the graph of f .

(c) Sketch the graph of f in the xy -plane provided below.

**Part A**

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f'(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y -axis then f is even

Unit 1 AP Topics 5-8

$$f(-x) = f(x) \quad \therefore \boxed{a = 0}$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y -axis then f is even

$$f(-x) = f(x) \quad \therefore \boxed{a = 0}$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$$

Part B

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

Unit 1 AP Topics 5-8VERTICAL: $x = 2$, $x = -2$ HORIZONTAL: $y = 0$ 

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

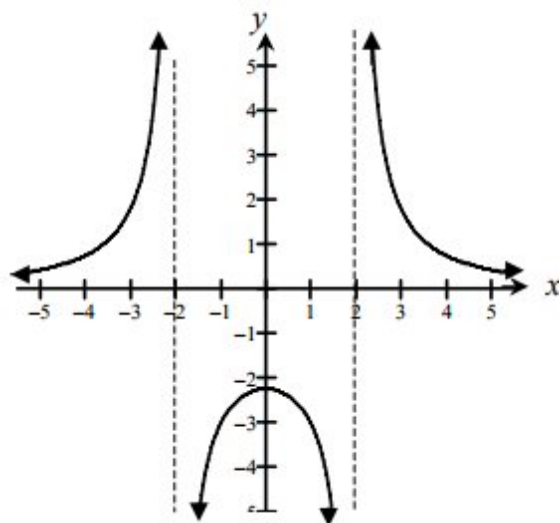
$$f(x) = \frac{9}{x^2 - 4}$$

VERTICAL: $x = 2$, $x = -2$ HORIZONTAL: $y = 0$ **Part C**

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y-int



Unit 1 AP Topics 5-8



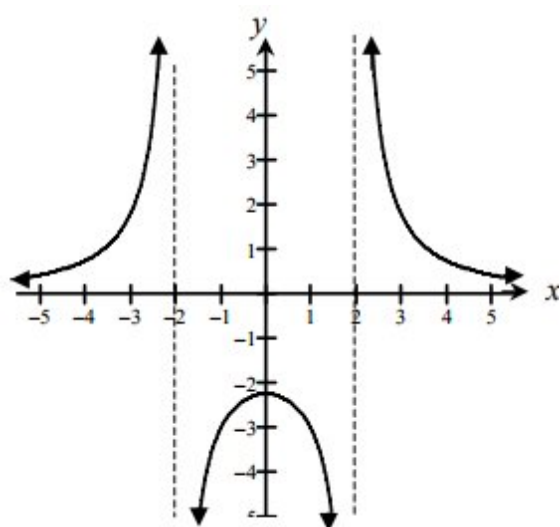
0	1	2	3
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The student response earns all of the following points:

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y -int



Unit 1 AP Topics 5-8

6. Let f be the function that is given by $f(x) = ax + b/x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y -axis.

(ii) $\lim_{x \rightarrow 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

Determine the values of a , b , and c .

Part A

1 point is earned for $a=0$

1 point is earned for $c=4$

1 point is earned for $f(x)$

1 point is earned for finding $b=5$

If graph symmetric to y -axis then f is even

$f(-x) = f(x)$ therefore $a = 0$

$\lim_{x \rightarrow 2^+} f(x) = +\infty$ therefore $c = 4$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \text{ therefore } b = 9$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $a=0$

1 point is earned for $c=4$

1 point is earned for $f(x)$

1 point is earned for finding $b=5$

If graph symmetric to y -axis then f is even

Unit 1 AP Topics 5-8

$$f(-x) = f(x) \text{ therefore } a = 0$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \text{ therefore } c = 4$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \text{ therefore } b = 9$$

Unit 1 AP Topics 5-8

7. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

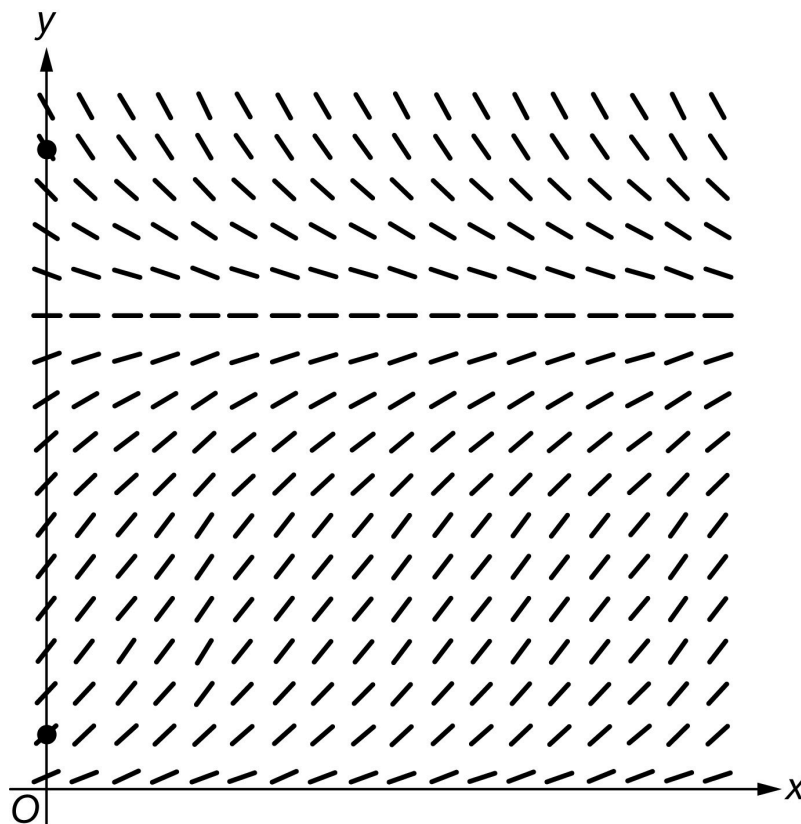
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right)$.

(a) A portion of the slope field for the differential equation is given below. Sketch the solution curves through the two indicated points for $x \geq 0$.



(b) Find $\lim_{y \rightarrow 50} \frac{dy}{dx}$.

(c) Let f be the particular solution to the differential equation with initial condition $f(0) = 20$. Find $f'(0)$, and explain why f is increasing for all x .

(d) For $0 < y < 40$, at what value of y does $\frac{dy}{dx}$ attain its maximum value? Justify your answer.

Unit 1 AP Topics 5-8

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Each solution curve must pass through the appropriate point, extend reasonably close to the right edge of the square, and have no obvious conflicts with the given slope lines.

Ignore any portion of a solution curve that extends beyond the square.

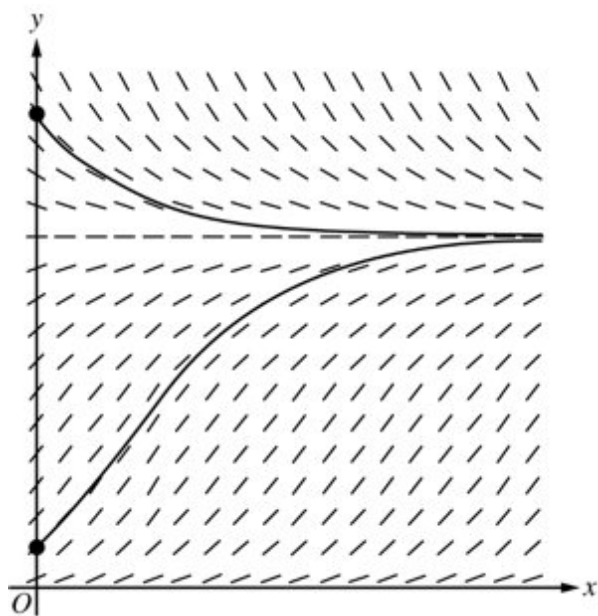


0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Top solution curve
- ☐ Bottom solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 0 earns the point.

The answer of $-\frac{50}{4} \ln\left(\frac{50}{50}\right)$ need not be simplified, but any subsequent simplification must be correct.

Limit notation is not required.

Unit 1 AP Topics 5-8

Expressions that equate the limit with the derivative such as $\lim_{y \rightarrow 50} \frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right) = \frac{-50}{4} \ln\left(\frac{50}{50}\right)$, or other linkage errors, will not earn the point, even in the presence of the correct evaluation.

Expressions with incomplete limit notation, such as $\lim_{y \rightarrow 50} = \frac{-50}{4} \ln\left(\frac{50}{50}\right)$, are eligible for the point.

The expression $\lim_{y \rightarrow 50} \frac{-50}{4} \ln\left(\frac{50}{50}\right)$ is not incorrect, but the limit must be resolved to earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Limit

Solution:

$$\lim_{y \rightarrow 50} \frac{dy}{dx} = \lim_{y \rightarrow 50} \frac{-y}{4} \ln\left(\frac{y}{50}\right) = \frac{-50}{4} \ln\left(\frac{50}{50}\right) = 0$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $-5 \ln\left(\frac{2}{5}\right)$ or the equivalent; no supporting work is necessary.

The evaluation does not need to be simplified, and incorrect attempts at simplifying a correct expression will not be read.

For the second point to be earned, the response must:

- Argue that $\frac{dy}{dx}$ is positive for $0 < y < 50$, thus f is increasing for $0 < y < 50$.
- Argue that the particular solution cannot cross the line $y = 50$. (This may be done implicitly by restricting the domain of consideration to $0 < y < 50$.)

Explanations may be graphical but must address both of the above points.

Explanations that appeal only to the sign of $f'(0)$ will not earn this point.



0	1	2
---	---	---

Unit 1 AP Topics 5-8

The student response accurately includes **both** of the criteria below.

- ☐ $f'(0)$
- ☐ Explanation

Solution:

$$f'(0) = \frac{dy}{dx} \Big|_{(x,y)=(0,20)} = \frac{-20}{4} \ln\left(\frac{20}{50}\right) = -5 \ln\left(\frac{2}{5}\right)$$

Because $\frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right) > 0$ for $0 < y < 50$, f is increasing provided $f(x) < 50$.

Note that $y = 50$ is also a solution to the differential equation, so the graph of the particular solution $y = f(x)$ with $f(0) = 20$ is below the line $y = 50$ and cannot cross the line $y = 50$.

Thus, f is increasing for all x .

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by correctly applying the product rule in finding $\frac{d^2y}{dx^2}$, with or without using the chain rule. For example, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y}$ would earn the first point.

The second point is earned for correctly applying the chain rule (twice) in finding $\frac{d^2y}{dx^2}$. This point can be earned even if an error was made in applying the product rule that resulted in the first point not being earned. In general, expressions of the form $\frac{d^2y}{dx^2} = c \cdot \ln\left(\frac{y}{50}\right) \cdot \frac{dy}{dx} + g(y) \cdot \frac{dy}{dx}$, where $g(y)$ may be a nonzero constant and c is a nonzero constant, earn this point.

For the first two points, if both the factors 50 and $\frac{1}{50}$ are not present in the response, assume that these factors were canceled. Furthermore, in the presence of the factor 50, assume that a missing factor $\frac{1}{50}$ is an error in the application of the chain rule. Thus, $\frac{d^2y}{dx^2} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{50}{y} \cdot \frac{dy}{dx}$ would earn the first point but not the second point.

For the first two points, each point is earned with the correct expression and any subsequent errors in simplification come off of the fourth point.

The third point is earned when the declared $\frac{d^2y}{dx^2}$ is set equal to zero and one initial (correct or otherwise) algebraic step towards solving this equation for y is made. However, the declared expression for $\frac{d^2y}{dx^2}$ must contain the variable y and must be distinct from $\frac{dy}{dx} = \frac{-y}{4} \ln\left(\frac{y}{50}\right)$. (This means a score of 0-0-1-0 is possible in this part.)

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The fourth point cannot be earned without having earned the first three points.

The fourth point requires a discussion of the behavior of $\frac{d^2y}{dx^2}$ both to the left and right of $y = 50e^{-1}$.

Sign charts, labeled or otherwise, are not read for the fourth point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Attempts to solve $\frac{d^2y}{dx^2} = 0$ for y
- ☐ Answer with justification

Solution:

$$\begin{aligned}\frac{d^2y}{dx^2} &= -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y} \cdot \frac{dy}{dx} \\ &= -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \ln\left(\frac{y}{50}\right) - \frac{y}{4} \cdot \frac{1}{y} \cdot \frac{dy}{dx} = -\frac{1}{4} \cdot \frac{dy}{dx} \cdot \left(\ln\left(\frac{y}{50}\right) + 1\right)\end{aligned}$$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow \left(\ln\left(\frac{y}{50}\right) + 1\right) = 0 \Rightarrow y = 50e^{-1}$$

Because $\frac{d^2y}{dx^2} > 0$ for $0 < y < 50e^{-1}$ and $\frac{d^2y}{dx^2} < 0$ for $50e^{-1} < y < 40$, $\frac{dy}{dx}$ attains its maximum value at $y = 50e^{-1}$.

8. $\lim_{x \rightarrow -7} \frac{x+7}{|x+7|}$ is

- (A) -1
- (B) 0
- (C) 1
- (D) nonexistent



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9. If $\int_1^x f(t)dt = \frac{20x}{\sqrt{4x^2+21}} - 4$, then $\int_1^\infty f(t)dt$ is

(A) 6



(B) 1

(C) -3

(D) -4

(E) divergent

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10. A tank contains 30 liters of water at time $t = 0$. For time $0 \leq t \leq 20$ minutes, water flows into the tank at a rate modeled by the function f defined by $f(t) = 2 \ln(t + 3) - 1.6$. For time $20 < t \leq 45$ minutes, water flows into the tank at a rate modeled by the function g defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

**Part A**

Find the rate of change of g at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

Part B

How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

Part C

Let w be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \leq t \leq 20 \\ g(t) & \text{for } 20 < t \leq 45 \end{cases}$. Find $\lim_{t \rightarrow 20^+} w(t)$. Is w continuous at $t = 20$? Use the definition of continuity to justify your answer.

Part D

Consider the function w defined in part C. Find the average value of w from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a correct value either rounded or truncated to three places after the decimal and for labeling the value as $g'(23)$. An inappropriately rounded answer does not earn the point.
- **P1** can also be earned for the exact value $\frac{-7.5}{23(\ln 23)^2}$.

Model Solution:

$$g'(23) = -0.033$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(23)$

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Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P2** is earned for correct units, whether or not they are attached to a numerical value for the rate of change.
- **P2** is also earned for the units “liters/minute².”

Model Solution:

The rate of change in the rate at time $t = 23$ is -0.033 liters per minute per minute.



0	1
---	---

The response accurately includes the criteria below.

- Units

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P3** is earned with a definite integral containing the correct integrand and correct lower and upper limits, with or without the differential.
- If the differential dt is missing:
 - Writing $\int_0^{20} f(t)$ or $30 + \int_0^{20} f(t)$ earns **P3**.
 - Because $\int_0^{20} f(t) + 30$ may mean $\int_0^{20} f(t)dt + 30$ or $\int_0^{20} (f(t) + 30)dt$:
 - Writing $\int_0^{20} f(t) + 30$ in the presence of a correct answer of 95.641 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of an answer of 95, 96, 95.6, or 95.64 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of no answer or an incorrect answer does not earn **P3**.
- Writing $\int f(t)dt$, $30 + \int f(t)dt$, $30 + \int f(t)$, or $\int f(t) + 30$ and never providing the definite integral, does not earn **P3**.

Model Solution:

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$$30 + \int_0^{20} f(t)dt = 30 + \int_0^{20} (2 \ln(t + 3) - 1.6)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P4** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P4** for the correct answer:
 - $\int_0^{20} f(t)dt = 65.641060$ so the number of liters of water is 95.641.
 - $\int_0^{20} f(t)dt = 65.641060 + 30 = 95.641060$ (Incorrect linkage is not considered in scoring.)
 - $\int_0^{20} f(t)dt = 95.641060$ (Incorrect linkage is not considered in scoring.)

Model Solution:

There are 95.641 liters of water in the tank at time $t = 20$ minutes.



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

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- To earn **P5**, a response must indicate $\lim_{t \rightarrow 20^+} w(t) = 4.704$ (or 4.703), either symbolically or verbally. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn **P5**, unless an earlier point was not earned due to inappropriate rounding.
 - A response of “the right-hand limit is 4.704” is sufficient to earn **P5**.

Model Solution:

$$\lim_{t \rightarrow 20^+} w(t) = \lim_{t \rightarrow 20^+} g(t) = 4.703562$$

The limit as t approaches 20 from the right is 4.704 (or 4.703).



0	1
---	---

The response accurately includes the criteria below.

- Right-hand limit

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P6**, a response must include $w(20) = 4.671$ (or 4.67 or 4.6), $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, and a conclusion of not continuous (with or without indicating $t = 20$).
- Alternate solution:

$$\lim_{t \rightarrow 20^-} w(t) = \lim_{t \rightarrow 20^-} f(t) = 4.670988$$

Because $\lim_{t \rightarrow 20^-} w(t) \neq \lim_{t \rightarrow 20^+} w(t)$, w is not continuous at $t = 20$.

Such a response is scored analogously to the model solution and earns **P6**.

- Special case: A response of “ $\lim_{t \rightarrow 20^+} w(t) = 4.7$ and $\lim_{t \rightarrow 20^-} w(t) = w(20) = 4.7$, therefore continuous” earns **P6**.

Model Solution:

$$w(20) = f(20) = 4.670988$$

Because $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, w is not continuous at $t = 20$.

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0	1
---	---

The response accurately includes the criteria below.

- Answer with justification

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P7**, a response must present one or more definite integrals with correct integrands, with or without differentials. Limits of integration are not considered in scoring for **P7**. That is, any of the following earns **P7**: $\int_a^c w(t)dt$, $\int_a^b w(t)dt + \int_b^c w(t)dt$, or $\int_a^b f(t)dt + \int_b^c g(t)dt$, where a , b , and c are real numbers between 0 and 45, inclusive.

Model Solution:

$$\frac{1}{30-15} \int_{15}^{30} w(t)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P8**, a response must present a correct definite integral and evidence of division by 15.
 - A response that earned **P7** earns **P8** for $\frac{1}{15} \int_{15}^{30} w(t)dt$, $\frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right)$, $\frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right)$, or equivalent.

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- Presenting the correct integral and the correct answer will suffice as evidence of division by 15 and will earn **P8**. Work presented between earning **P7** and the correct answer that does not reveal incorrect work done in the calculator will be treated as scratch work and will not be considered in scoring.
 - “ $\int_{15}^{30} w(t)dt = 67.559$, so 4.504” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer.
 - “ $\int_{15}^{30} w(t)dt = \frac{67.559}{15} = 4.504$ ” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer. The incorrect communication between the integral and the answer does not reveal incorrect work done in the calculator. This work is treated as scratch work and is not considered in scoring of **P8**.
- A response that did not earn **P7** is eligible to earn **P8** for $\frac{1}{15} \int_{15}^{30} f(t)dt$ or $\frac{1}{15} \int_{15}^{30} g(t)dt$.

Model Solution:

$$\begin{aligned} & \frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right) \\ &= \frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right) \end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Average value formula

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The average value of w from time $t = 15$ and time $t = 30$ minutes is 4.504 (or 4.503) liters per minute.



0	1
---	---

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The response accurately includes the criteria below.

- Answer

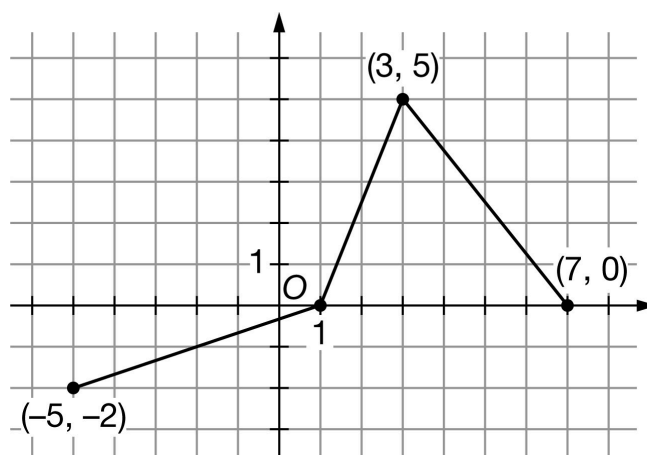
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11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-5 \leq x \leq 7$ consists of three line segments, as shown in the figure above. It is known that $h(-5) = 6$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 7$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 7$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 1$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(3) = 5$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 1$ only

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· $x = 1$ and any additional x -coordinates not in $-5 < x < 7$

Does not earn the point:

· $x = 1$ and any additional x -coordinates in $-5 < x < 7$

· $x = 1$ with any incorrect supporting work

Do not read work related to additional values not in $-5 < x < 7$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 1$
- the slope of h changes from negative to positive at $x = 1$
- the graph changes from negative to positive at $x = 1$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 1$ and increasing to the right of $x = 1$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 1$

Note: If additional x -coordinates in $-5 < x < 7$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-5 < x < 7$ are identified in part (a), do not read these responses. Only read for values in the interval.

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0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -5$ and $x = 7$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-5) = 6$ in calculation for $h(1)$ or $h(7)$

Third point: Answer with justification

The response must declare as an answer the absolute maximum value of 15.

The ordered pair $(7, 15)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-5	6
1	0
7	15



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

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- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-5 < x < 7$ is a relative minimum, so the absolute maximum value of h on $-5 \leq x \leq 7$ must occur at an endpoint.

$$h(-5) = 6$$

$$\begin{aligned} h(7) &= h(-5) + \int_{-5}^7 h'(t) \, dt \\ &= 6 + \int_{-5}^1 h'(t) \, dt + \int_1^7 h'(t) \, dt \\ &= 6 - \frac{1}{2} \cdot 6 \cdot 2 + \frac{1}{2} \cdot 6 \cdot 5 = 15 \end{aligned}$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 7$ is $h(7) = 15$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from increasing to decreasing
- h'' changes from positive to negative at $x = 3$
- the slope of h' changes from positive to negative at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative maximum exists on the graph of h' at $x = 2$

Does not earn the point:

- the concavity changes at $x = 3$ [Must appeal to the change in sign of h'']
- h' changes from positive to negative
- the graph changes from positive to negative
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of h' at $x = 2$

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- the acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from increasing to decreasing there .

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 1$ is not the location of a point of inflection because

- h' is increasing for $x < 1$ and $x > 1$
- h' is increasing through $x = 1$
- h'' is always positive before and after $x = 1$
- the slope of h' is always positive before and after $x = 1$
- h'' does not change sign at $x = 1$
- a relative extrema does not exist on the graph of h' at $x = 1$

Does not earn the point:

- the concavity does not change [Must appeal to the change in sign of h'']
- h' is always positive
- h' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

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Solution:

$x = 1$ is not the location of a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow -2} (h'(x) + 1) = 0 \text{ and } \lim_{x \rightarrow -2} (x^2 - 4) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x)}{2x}$$

$$\cdot \frac{h'(-2)}{2(-2)}$$

$$\cdot \frac{h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{12}$ or equivalent

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0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$ and $\lim_{x \rightarrow -2} h''(x) = h''(-2)$.

$$\lim_{x \rightarrow -2} h'(x) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \lim_{x \rightarrow -2} \frac{h''(x)}{2x}$$

$$= \frac{h''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{3}}{-4} = -\frac{1}{12}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

1st point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(3) = 3 \cdot h'(3) + h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the 1st point

Must be our answer, $g'(3) = 20$ or equivalent

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0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(3) = 3 \cdot h'(3) + h(3) = 3 \cdot 5 + 5 = 20$$

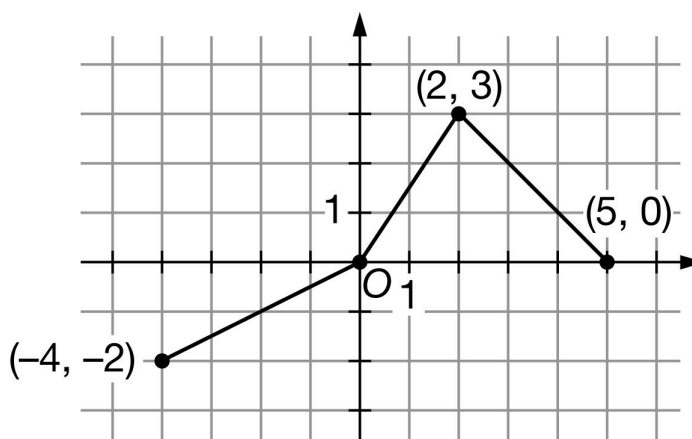
Unit 1 AP Topics 5-8

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-4 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-4) = 7$.

- Find the x -coordinate of each critical point of g on the open interval $-4 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute maximum value of g on the interval $-4 \leq x \leq 5$. Justify your answer.
- Is $x = 2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = x \cdot g(x)$. It is known that $g(2) = 6$. Find the value of $h'(2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

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- $x = 0$ and any additional x -coordinates not in $-4 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-4 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-4 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $g'(x)$ changes from negative to positive at $x = 0$
- the slope of g changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- g is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-4 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is NOT eligible to earn the point.

Note: If additional x -coordinates not in $-4 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.

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0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for g because $g'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -4$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-4) = 7$ in a calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 10.5.

The ordered pair $(5, 10.5)$ is not a declaration of the maximum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-4	7
0	3
6	10.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

Unit 1 AP Topics 5-8

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-4 < x < 5$ is a relative minimum, so the absolute maximum value of g on $-4 \leq x \leq 5$ must occur at an endpoint.

$$g(-4) = 7$$

$$\begin{aligned} g(5) &= g(-4) + \int_{-4}^5 g'(t) \, dt \\ &= 7 + \int_{-4}^0 g'(t) \, dt + \int_0^5 g'(t) \, dt \\ &= 7 - \frac{1}{2} \cdot 4 \cdot 2 + \frac{1}{2} \cdot 3 \cdot 5 = 10.5 \end{aligned}$$

Therefore, the absolute maximum value of g on $-4 \leq x \leq 5$ is $g(5) = 10.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = 2$
- the slope of g' changes from positive to negative at $x = 2$
- g'' changes sign at $x = 2$
- a relative extrema exists on the graph of g' at $x = 2$
- a relative maximum exists on the graph of g' at $x = 2$

Does not earn the point:

- the concavity at $x = 2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ or $g''(x)$ does not exist (DNE)
- a relative minimum exists on the graph of g' at $x = 2$

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- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = 2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is increasing for $x < 0$ and $x > 0$
- g' is increasing through $x = 0$
- g'' is always positive before and after $x = 0$
- the slope of g' is always positive before and after $x = 0$
- g'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always positive
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

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Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is increasing on both sides of $x = 0$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow -2} (g'(x) + 1) = 0 \text{ and } \lim_{x \rightarrow -2} (x^2 - 4) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x)}{2x}$$

$$\cdot \frac{g'(-2)}{2(-2)}$$

$$\cdot \frac{g'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{8}$ or equivalent

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0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} g'(x) = g'(-2)$ and $\lim_{x \rightarrow -2} g''(x) = g''(-2)$.

$$\lim_{x \rightarrow -2} (g'(x) + 1) = g'(-2) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} &= \lim_{x \rightarrow -2} \frac{g''(x)}{2x} \\ &= \frac{g''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{2}}{-4} = -\frac{1}{8} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product rule

Applies product rule: $h'(x) = x \cdot g'(x) + g(x)$

Product rule may appear as $h'(2) = 2 \cdot g'(2) + g(2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(2) = 12$ or equivalent

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0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$h'(x) = x \cdot g'(x) + g(x)$$

$$h'(2) = 2 \cdot g'(2) + g(2) = 2 \cdot 3 + 6 = 12$$

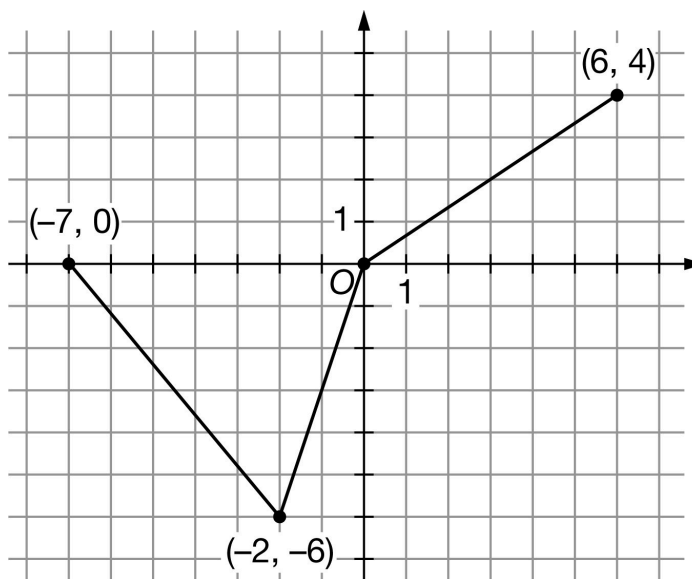
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13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. It is known that $h(-7) = 2$.

- Find the x -coordinate of each critical point of h on the open interval $-7 < x < 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-7 \leq x \leq 6$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow 3} \frac{2h'(x)-4}{x^2-9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(-2) = -13$. Find the value of $g'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-7 < x < 6$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-7 < x < 6$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-7 < x < 6$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 0$
- the slope of h changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$
- Any incorrect argument

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Note: If additional x -coordinates in $-7 < x < 6$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-7 < x < 6$ are identified in part (a), do not read reasons for these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -7$ and $x = 6$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-7) = 2$ in a calculation for $h(0)$ and/or $h(6)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 2.

The ordered pair $(-7, 2)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-7	2
0	-19
6	-7

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0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-7 < x < 6$ is a relative minimum, so the absolute maximum value of h on $-7 \leq x \leq 6$ must occur at an endpoint.

$$h(-7) = 2$$

$$\begin{aligned} h(6) &= h(-7) + \int_{-7}^6 h'(t) dt \\ &= 2 + \int_{-7}^0 h'(t) dt + \int_0^6 h'(t) dt \\ &= 2 - \frac{1}{2} \cdot 7 \cdot 6 + \frac{1}{2} \cdot 6 \cdot 4 = -7 \end{aligned}$$

Therefore, the absolute maximum value of h on $-7 \leq x \leq 6$ is $h(-7) = 2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' changes from negative to positive at $x = -2$
- the slope of h' changes from negative to positive at $x = -2$
- h'' changes sign at $x = -2$
- a relative extrema exists on the graph of h' at $x = -2$
- a relative minimum exists on the graph of h' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of h'']

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- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = -2$
- The acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is increasing for $x < 0$ and $x > 0$
- h' is increasing through $x = 0$
- h'' is always positive before and after $x = 0$
- the slope of h' is always positive before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of h'']
- h' is always positive
- h' does not change sign

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0

1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow 3} (2h'(x) - 4) = 0 \text{ and } \lim_{x \rightarrow 3} (x^2 - 9) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} \rightarrow 0$$

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x)}{2x}$$

$$\cdot \frac{2h''(3)}{2(3)}$$

$$\cdot \frac{2h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

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Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{2}{9}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = 3$. $\Rightarrow \lim_{x \rightarrow 3} h'(x) = h'(3)$ and $\lim_{x \rightarrow 3} h''(x) = h''(3)$.

$$\lim_{x \rightarrow 3} 2h'(x) - 4 = 2 \cdot 2 - 4 = 0$$

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \lim_{x \rightarrow 3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(3)}{3} = \frac{\frac{2}{3}}{3} = \frac{2}{9}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(-2) = -2 \cdot h'(-2) + h(-2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

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Eligibility: Earns the first point

Must be our answer, $g'(-2) = -1$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(-2) = -2 \cdot h'(-2) + h(-2) = -2 \cdot (-6) + (-13) = -1$$

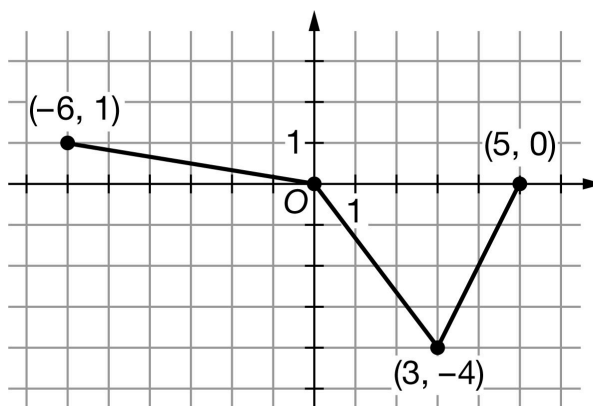
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14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $h(-6) = 5$.

- Find the x -coordinate of each critical point of h on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute minimum value of h on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -3} \frac{2h'(x)-1}{x^2-9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = 2x \cdot h(x)$. It is known that $h(3) = 2$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

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· $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

· $x = 0$ and any additional x -coordinates in $-6 < x < 5$

· $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $h'(x)$ changes from positive to negative at $x = 0$
- the slope of h changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- h is increasing to the left of $x = 0$
- and decreasing to the right of $x = 0$
- (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to

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earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for h because $h'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the 1st point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-6) = 5$ in a calculation for $h(0)$ or $h(5)$

Third point: Answer with justification

The response must declare as an answer the minimum value of -2 .

The ordered pair $(5, -2)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-6	5
0	8
5	-2



0	1	2	3
---	---	---	---

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The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of h on $-6 \leq x \leq 5$ must occur at an endpoint.

$$h(-6) = 5$$

$$\begin{aligned} h(5) &= h(-6) + \int_{-6}^5 h'(t) dt \\ &= 5 + \int_{-6}^0 h'(t) dt + \int_0^5 h'(t) dt \\ &= 5 + \frac{1}{2} \cdot 6 \cdot 1 - \frac{1}{2} \cdot 5 \cdot 4 = -2 \end{aligned}$$

Therefore, the absolute minimum value of h on $-6 \leq x \leq 5$ is $h(5) = -2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' shifts from negative to positive at $x = 3$
- the slope of h' shifts from negative to positive at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative minimum exists on the graph of h' at $x = 3$

Does not earn the point:

- the concavity at $x = 3$ changes. [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive

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- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = 3$
- the acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is decreasing for $x < 0$ and $x > 0$
- h' is decreasing through $x = 0$
- h'' is always negative before and after $x = 0$
- the slope of h' is always negative before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change. [Must appeal to the lack of change in sign of h'']
- h' is always negative
- h' does not change sign



0	1
---	---

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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of h because h' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -3$

Earns the point:

$$\cdot \lim_{x \rightarrow -3} (2h'(x) - 1) = 0 \text{ and } \lim_{x \rightarrow -3} (x^2 - 9) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -3} \frac{2h'(x)}{2x}$$

$$\cdot \frac{2h'(-3)}{2(-3)}$$

$$\cdot \frac{2h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Unit 1 AP Topics 5-8

Must be our answer $\frac{1}{18}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -3$. $\Rightarrow \lim_{x \rightarrow -3} h'(x) = h'(-3)$ and $\lim_{x \rightarrow -3} h''(x) = h''(-3)$.

$$\lim_{x \rightarrow -3} 2h'(x) - 1 = 2 \cdot \frac{1}{2} - 1 = 0$$

$$\lim_{x \rightarrow -3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \lim_{x \rightarrow -3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(-3)}{-3} = \frac{-\frac{1}{6}}{-3} = \frac{1}{18}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = 2x \cdot h'(x) + 2 \cdot h(x)$

Product rule may appear as $g'(3) = 6 \cdot h'(3) + 2 \cdot h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Unit 1 AP Topics 5-8

Must be our answer, $g'(3) = -20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$g'(x) = 2x \cdot h'(x) + 2h(x)$$

$$g'(3) = 2 \cdot 3 \cdot h'(3) + 2h(3) = 6 \cdot (-4) + 2 \cdot 2 = -20$$

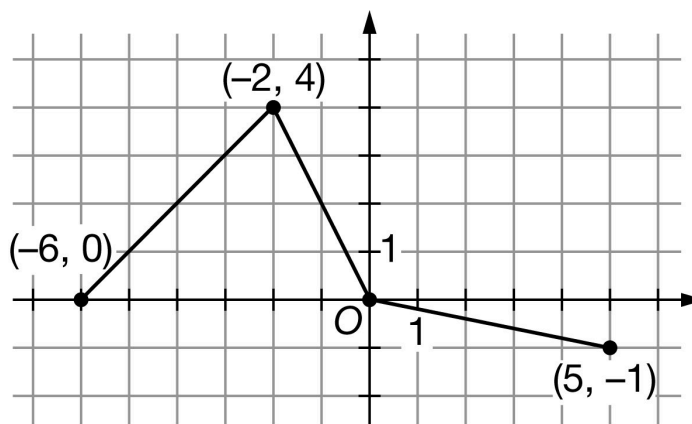
Unit 1 AP Topics 5-8

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-6) = -1$.

- Find the x -coordinate of each critical point of g on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute minimum value of g on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -1} \frac{g'(x)-2}{x^2-1}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = 3x \cdot g(x)$. It is known that $g(-2) = 7$. Find the value of $h'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

Unit 1 AP Topics 5-8

· $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

· $x = 0$ and any additional x -coordinates in $-6 < x < 5$

· $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $g'(x)$ changes from positive to negative at $x = 0$
- the slope of g changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- g is increasing to the left of $x = 0$ and decreasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval

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0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for g because $g'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-6) = -1$ in calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the absolute minimum value of -1 .

The ordered pair $(-6, -1)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-6	-1
0	11
5	8.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

Unit 1 AP Topics 5-8

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of g on $-6 \leq x \leq 5$ must occur at an endpoint.

$$g(-6) = -1$$

$$\begin{aligned} g(5) &= g(-6) + \int_{-6}^5 g'(t) dt \\ &= -1 + \int_{-6}^0 g'(t) dt + \int_0^5 g'(t) dt \\ &= -1 + \frac{1}{2} \cdot 6 \cdot 4 - \frac{1}{2} \cdot 5 \cdot 1 = \frac{17}{2} \end{aligned}$$

Therefore, the absolute minimum value of g on $-6 \leq x \leq 5$ is $g(-6) = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = -2$
- the slope of g' changes from positive to negative at $x = -2$
- g'' changes sign at $x = -2$
- a relative extrema exists on the graph of g' at $x = -2$
- a relative maximum exists on the graph of g' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ OR $g''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of g' at $x = -2$

Unit 1 AP Topics 5-8

- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is decreasing for $x < 0$ and $x > 0$
- g' is decreasing through $x = 0$
- g'' is always negative before and after $x = 0$
- the slope of g' is always negative before and after $x = 0$
- g'' does not change sign at $x = 0$.
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always negative
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

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Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -1$

Earns the point:

$$\cdot \lim_{x \rightarrow -1} (g'(x) - 2) = 0 \text{ and } \lim_{x \rightarrow -1} (x^2 - 1) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x)}{2x}$$

$$\cdot \frac{g''(-1)}{2(-1)}$$

$$\cdot \frac{g'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer 1 or equivalent

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0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -1$. $\Rightarrow \lim_{x \rightarrow -1} g'(x) = g'(-1)$ and $\lim_{x \rightarrow -1} g''(x) = g''(-1)$.

$$\lim_{x \rightarrow -1} g'(x) - 2 = 2 - 2 = 0$$

$$\lim_{x \rightarrow -1} (x^2 - 1) = 1^2 - 1 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} &= \lim_{x \rightarrow -1} \frac{g''(x)}{2x} \\ &= \frac{g''(-1)}{2 \cdot (-1)} = \frac{-2}{-2} = 1 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $h'(x) = 3x \cdot g'(x) + 3 \cdot g(x)$

Product rule may appear as $h'(-2) = -6 \cdot g'(-2) + 3 \cdot g(-2)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(-2) = -3$ or equivalent

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0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$h'(x) = 3x \cdot g'(x) + 3g(x)$$

$$h'(-2) = 3 \cdot (-2) \cdot g'(-2) + 3g(-2) = -6 \cdot 4 + 3 \cdot 7 = -3$$

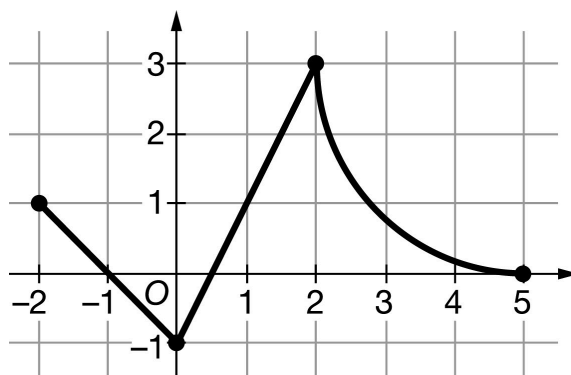
Unit 1 AP Topics 5-8

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $-6 \leq x \leq 5$. The figure above shows a portion of the graph of f , consisting of two line segments and a quarter of a circle centered at the point $(5, 3)$. It is known that the point $(3, 3 - \sqrt{5})$ is on the graph of f .

- If $\int_{-6}^5 f(x) \, dx = 7$, find the value of $\int_{-6}^{-2} f(x) \, dx$. Show the work that leads to your answer.
- Evaluate $\int_3^5 (2f'(x) + 4) \, dx$.
- The function g is given by $g(x) = \int_{-2}^x f(t) \, dt$. Find the absolute maximum value of g on the interval $-2 \leq x \leq 5$. Justify your answer.
- Find $\lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for showing evidence of using the property of adding integrals. This point can be earned symbolically or numerically with correct values of the integrals.

Simplification is not required to earn the second point or the third point

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0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\int_{-6}^5 f(x) dx = \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx$ integral
- ☐ $\int_{-2}^5 f(x) dx$
- ☐ answer

Solution:

$$\begin{aligned}\int_{-6}^5 f(x) dx &= \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx \\ \Rightarrow 7 &= \int_{-6}^{-2} f(x) dx + 2 + \left(9 - \frac{9\pi}{4}\right) \\ \Rightarrow \int_{-6}^{-2} f(x) dx &= 7 - \left(11 - \frac{9\pi}{4}\right) = \frac{9\pi}{4} - 4\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an antiderivative of $f'(x)$ and use of the limits of integration. Responses with errors in the integrand, such as a missing coefficient, earn the first point; such errors are considered for the second point.

Simplification is not required to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ answer

Solution:

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$$\begin{aligned}
 \int_3^5 (2f'(x) + 4) \, dx &= 2 \int_3^5 f'(x) \, dx + \int_3^5 4 \, dx \\
 &= 2(f(5) - f(3)) + 4(5 - 3) \\
 &= 2\left(0 - \left(3 - \sqrt{5}\right)\right) + 8 \\
 &= 2(-3 + \sqrt{5}) + 8 = 2 + 2\sqrt{5}
 \end{aligned}$$

OR

$$\begin{aligned}
 \int_3^5 (2f'(x) + 4) \, dx &= [2f(x) + 4x]_{x=3}^{x=5} \\
 &= (2f(5) + 20) - (2f(3) + 12) \\
 &= (2 \cdot 0 + 20) - \left(2\left(3 - \sqrt{5}\right) + 12\right) \\
 &= 2 + 2\sqrt{5}
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for connecting $g'(x) = f(x)$ analytically or verbally.

The second point is earned with identification of $x = -1$ regardless of what other candidates are considered.

The third point is earned for calculating $g(-1)$ and $g(5)$, declaring the absolute maximum to be $11 - \frac{9\pi}{4}$, and correctly reasoning for excluding $g(-2)$. Note that the ordered pair $\left(5, 11 - \frac{9\pi}{4}\right)$ by itself is not a declaration of the absolute maximum.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ identifies $x = -1$ as a candidate
- ☐ answer with justification

Solution:

$$g'(x) = f(x) = 0 \Rightarrow x = -1, x = \frac{1}{2}, x = 5$$

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x	$g(x)$
-2	0
-1	$\frac{1}{2}$
$\frac{1}{2}$	$-\frac{1}{4}$
5	$11 - \frac{9\pi}{4}$

On the interval $-2 \leq x \leq 5$, the absolute maximum value of g is $g(5) = 11 - \frac{9\pi}{4}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer. Simplification, including $\arctan 1 = \frac{\pi}{4}$, is not required.



0	1
---	---

The student response accurately includes the criteria below.

☐ · answer

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x} &= \frac{10^1 - 3f'(1)}{f(1) - \arctan 1} \\ &= \frac{10 - 3 \cdot 2}{1 - \arctan 1} = \frac{4}{1 - \frac{\pi}{4}} \end{aligned}$$

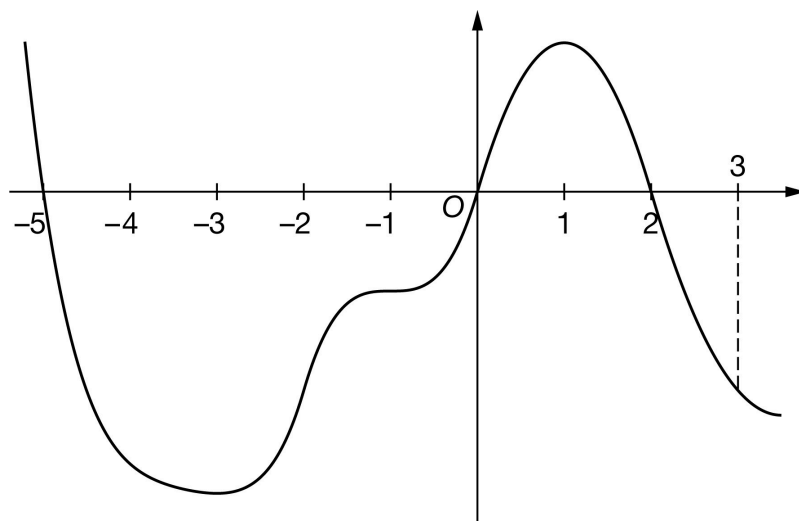
Unit 1 AP Topics 5-8

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-5.5 \leq x \leq 3.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-5, 0]$, $[0, 2]$, and $[2, 3]$ are 57, 12, and 7, respectively. The graph of f has horizontal tangents at $x = -3$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_2^x f(t) \, dt$ for $-5.5 \leq x \leq 3.5$.

- Find the x -coordinate of each critical point of g on the interval $-5.5 < x < 3.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of g on the interval $-5.5 < x < 3.5$. Justify your answer.
- Find the value of $g(3)$.
- Find the value of $g(-5)$. Show the computations that lead to your answer.
- Must there exist a value of k , for $-5 < k < 0$, such that $g'(k)$ is equal to the average rate of change of g over the interval $-5 \leq x \leq 0$? Justify your answer.
- Find $\lim_{x \rightarrow 3} \frac{5g(x)}{2x}$. Show the computations that lead to your answer.
- The function r is defined by $r(x) = x^2 \cdot g(x)$. It is known that $f(3) = -12$. Find $r'(3)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -5$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5.5, 3.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5.5, 3.5)$. The point is still earned provided the three critical points, $x = -5$, $x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5, 0)$, $(0, 0)$, $(2, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -5$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -5$, g has a relative maximum) is eligible for at most one of the three points.

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Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-5)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-5.5) < 45$
- $h(-5) = 45$
- $h(0) = -12$
- $h(2) = 0$
- $h(3) = -7$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -5$, g has a relative maximum because $g' = f$ is positive for $x < -5$ and $g' = f$ is negative for $x > -5$.
- At $x = 0$, g has a relative minimum because $g' = f$ is negative for $x < 0$ and $g'(x) = f(x)$ is positive for $x > 0$.
- At $x = 2$, g has a relative maximum because $g' = f$ is positive for $x < 2$ and $g' = f$ is negative for $x > 2$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Unit 1 AP Topics 5-8

☐ Relative maximum with justification

Solution:

At $x = -5$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 0$, g Alt text: g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

At $x = 2$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g' is eligible for all three points.

If a response gives justifications based only on f , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$.

Justification for the first two points should reference one of the following:

- $g'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $g'(x)$
- $g''(x)$ changes sign
- $f'(x)$ changes sign

“ $g'(x) = f(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to f as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

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If $x = -3$ and $x = 1$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -3$ with justification
- ☐ $x = 1$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -3$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from decreasing to increasing there.

At $x = 1$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -7 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g(3) = \int_2^3 f(t) dt = -7$$

Part E

Unit 1 AP Topics 5-8

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 57 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 57) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 57 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-57) - (-12)$ earns both points because numerical expressions need not be simplified.
- The expression $(-57) - 12$ earns the first point, but not the second point.
- The expression $-57 - (-12) - 7$ earns neither point.
- The expression $\int_2^{-5} f(t) dt = 45$ earns neither point.
- The expression $-\int_{-5}^0 f(t) dt - \int_0^2 f(t) dt = 45$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 57 and ± 12 , but it does earn the second point.
- $-57, -12$ earns neither point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-5}^0 f(t) dt \right| = 57$ and $\left| \int_2^0 f(t) dt \right| = 12$

☐ Answer

Solution:

$$\begin{aligned}
 g(-5) &= \int_2^{-5} f(t) dt = - \int_{-5}^2 f(t) dt \\
 &= - \int_{-5}^0 f(t) dt - \int_0^2 f(t) dt = -(-57) - 12 = 45
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-5, 0)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(k)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(k)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$g' = f \Rightarrow g \text{ is differentiable on } -5 \leq x \leq 0. \Rightarrow g \text{ is continuous on } -5 \leq x \leq 0.$$

Therefore, the Mean Value Theorem can be applied to g on the interval $-5 \leq x \leq 0$ to guarantee that there exists a value of k , for $-5 < k < 0$, such that $g'(k)$ equals the average rate of change of g over the interval $-5 \leq x \leq 0$.

Part G

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the first point include:

$$\cdot \frac{5(-7)}{2 \cdot 3}$$

$$\cdot -\frac{35}{6}$$

Responses that do not earn the first point include:

$$\cdot \frac{5 \cdot g(3)}{6}$$

$$\cdot \frac{5 \cdot g(3)}{2 \cdot 3}$$

Note: an incorrect value for $g(3)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g' = f \Rightarrow g \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 3} g(x) = g(3)$$

$$\lim_{x \rightarrow 3} \frac{5g(x)}{2x} = \frac{5g(3)}{2 \cdot (3)} = \frac{5(-7)}{6} = -\frac{35}{6}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of r that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or f).

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Fundamental Theorem of Calculus is evidenced by linking g' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(3)$. Since $f(3) = -12$ is given in the question, replacing $g'(3)$ directly with -12 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(3)$ that is brought in from part (d) can earn the third point.

$9(-12) + 6(-7)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a , b , c , or d is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$r'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$$

$$= x^2 \cdot f(x) + 2x \cdot g(x)$$

$$r'(3) = 3^2 \cdot f(3) + 2 \cdot 3 \cdot g(3) = 9 \cdot (-12) + 6 \cdot (-7) = -150$$

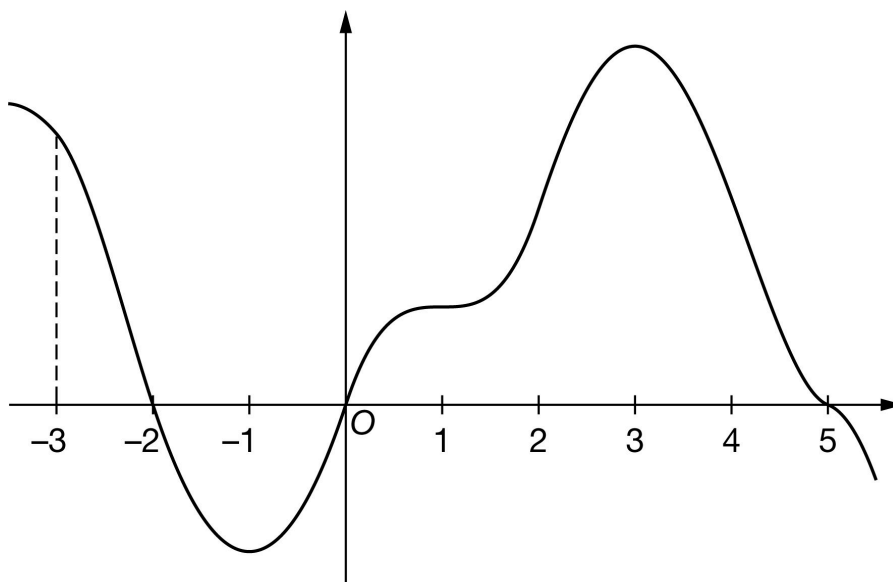
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18. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-3, -2]$, $[-2, 0]$, and $[0, 5]$ are 10, 13, and 59, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_0^x g(t) \, dt$ for $-3.5 \leq x \leq 5.5$.

- (a) Find the x -coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.
- (b) Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- (c) Find the x -coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.
- (d) Find the value of $h(5)$.
- (e) Find the value of $h(-3)$. Show the computations that lead to your answer.
- (f) Must there exist a value of p , for $-3 < p < 2$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.
- (g) Find $\lim_{x \rightarrow -3} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

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(h) The function f is defined by $f(x) = x \cdot h(x)$. It is known that $g(-3) = 18$. Find $f'(-3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -2$, $x = 0$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, 5.5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -2$, $x = 0$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -2$, h has a relative maximum) is

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eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-2)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-3) = 3$
- $h(-2) = 13$
- $h(0) = 0$
- $h(5) = 59$
- $h(5.5) < 59$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -2$, h has a relative maximum because $h' = g$ is positive for $x < -2$ and $h' = g$ is negative for $x > -2$.
- At $x = 0$, h has a relative minimum because $h' = g$ is negative for $x < 0$ and $h' = g$ is positive for $x > 0$.
- At $x = 5$, h has a relative maximum because $h' = g$ is positive for $x < 5$ and $h' = g$ is negative for $x > 5$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification

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- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 0$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only on h' is eligible for all three points.

If a response gives justifications based only on g , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

- $h'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $h'(x)$
- $h''(x)$ changes sign
- $g'(x)$ changes sign

“ $h'(x) = g(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to g as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be

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penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 59 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(5) = \int_0^5 g(t) dt = 59$$

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Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 10 and ± 13 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 10) \pm (\pm 13)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 13 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-10 - (-13)$ earns both points because numerical expressions need not be simplified.
- The expression $(-10) - 13$ earns the first point, but not the second point.
- The expression $-10 - (-13) - 3$ earns neither point.
- The expression $\int_0^{-3} g(t) dt = 3$ earns neither point.
- The expression $-\int_{-3}^{-2} g(t) dt - \int_{-2}^0 g(t) dt = 3$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 10 and ± 13 , but it does earn the second point.
- $-10, 13$ earns neither point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_{-2}^{-3} g(t) dt \right| = 10$ and $\left| \int_{-2}^0 g(t) dt \right| = 13$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(-3) &= \int_0^{-3} g(t) dt = - \int_{-3}^0 g(t) dt \\
 &= - \int_{-3}^{-2} g(t) dt - \int_{-2}^0 g(t) dt = -10 - (-13) = 3
 \end{aligned}$$

Part F

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = g \Rightarrow h \text{ is differentiable on } -3 \leq x \leq 2. \Rightarrow h \text{ is continuous on } -3 \leq x \leq 2.$$

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of p , for $-3 < p < 2$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

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Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{h(-3)}{-6} = \frac{3}{-6}$$

$$\cdot \frac{3}{-6}$$

Responses that do not earn the point are:

$$\cdot \frac{h(-3)}{-6}$$

$$\cdot \frac{h(-3)}{2 \cdot (-3)}$$

Note: an incorrect value for $h(-3)$ can be brought in from part (e) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -3} h(x) = h(-3)$$

$$\lim_{x \rightarrow -3} \frac{h(x)}{2x} = \frac{h(-3)}{2 \cdot (-3)} = \frac{3}{-6} = -\frac{1}{2}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of f that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(-3)$. Since $g(-3) = 18$ is given in the question, replacing $h'(-3)$ directly with 18 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(-3)$ that is brought in from part (e) can earn the third point.

$-3 \cdot 18 + 3$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a , b , or c is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = x \cdot h'(x) + 1 \cdot h(x)$$

$$= x \cdot g(x) + h(x)$$

$$f'(-3) = -3 \cdot g(-3) + h(-3) = -3 \cdot (18) + 3 = -51$$

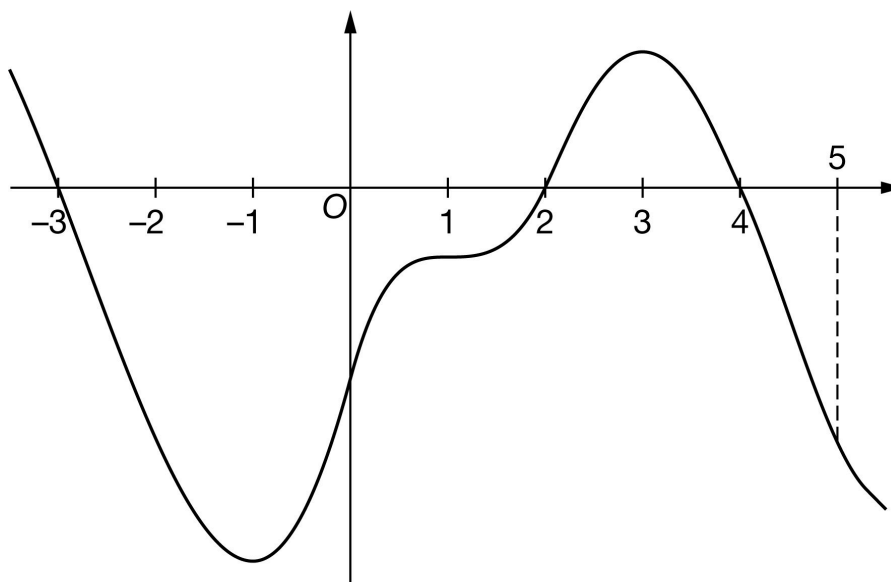
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19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the differentiable function h with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of h on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 58, 11, and 8, respectively. The graph of h has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_4^x h(t) \, dt$ for $-3.5 \leq x \leq 5.5$.

- Find the x -coordinate of each critical point of g on the interval $-3.5 < x < 5.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of g on the interval $-3.5 < x < 5.5$. Justify your answer.
- Find the value of $g(5)$.
- Find the value of $g(-3)$. Show the computations that lead to your answer.
- Must there exist a value of b , for $-3 < b < 2$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 2$? Justify your answer.
- Find $\lim_{x \rightarrow 5} \frac{g(x)}{3x}$. Show the computations that lead to your answer.

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(h) The function a is defined by $a(x) = x^2 \cdot g(x)$. It is known that $h(5) = -16$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, 5.5)$. The point is still earned provided the three critical points, $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 2$, h has a relative minimum) is eligible

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for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $g(-3.5) < 47$
- $g(-3) = 47$
- $g(2) = -11$
- $g(4) = 0$
- $g(5) = -8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -3$, g has a relative maximum because $g' = h$ is positive for $x < -3$ and $g' = h$ is negative for $x > -3$.
- At $x = 2$, g has a relative minimum because $g' = h$ is negative for $x < 2$ and $g' = h$ is positive for $x > 2$.
- At $x = 4$, g has a relative maximum because $g' = h$ is positive for $x < 4$ and $g' = h$ is negative for $x > 4$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification

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- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

At $x = 2$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 4$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only on g is eligible for all three points.

If a response gives justifications based only on h , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$.

Justification for the first two points should reference one of the following:

- $g'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $g'(x)$
- $g''(x)$ changes sign
- $h'(x)$ changes sign

“ $g'(x) = h(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to h as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized

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twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -8 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g(5) = \int_4^5 h(t) dt = -8$$

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Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 58 and ± 11 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 58) \pm (\pm 11)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values. The ± 58 and ± 11 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-58) - 11$ earns both points because numerical expressions need not be simplified.
- The expression $(-58) - 11$ earns the first point, but not the second point.
- The expression $-58 - 11 - 8$ earns neither point.
- The expression $\int_4^{-3} h(t) dt = 47$ earns neither point.
- The expression $-\int_{-3}^2 h(t) dt - \int_2^4 h(t) dt = 47$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 58 and ± 11 , but it does earn the second point.
- $-58, 11$ earns neither point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-3}^2 h(t) dt \right| = 58 \text{ and } \left| \int_4^2 h(t) dt \right| = 11$

☐ Answer

Solution:

$$\begin{aligned}
 g(-3) &= \int_4^{-3} h(t) dt = - \int_{-3}^4 h(t) dt \\
 &= - \int_{-3}^2 h(t) dt - \int_2^4 h(t) dt = -(-58) - 11 = 47
 \end{aligned}$$

Part F

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does the earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$g' = h \Rightarrow g \text{ is differentiable on } -3 \leq x \leq 2. \Rightarrow g \text{ is continuous on } -3 \leq x \leq 2.$$

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of b , for $-3 < b < 2$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 2$.

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Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{g(5)}{15} = \frac{-8}{15}$$

$$\cdot \frac{-8}{15}$$

Responses that do not earn the point are:

$$\cdot \frac{g(5)}{15}$$

$$\cdot \frac{g(5)}{3 \cdot 5}$$

Note: an incorrect value for $g(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g' = h \Rightarrow g \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 5} g(x) = g(5)$$

$$\lim_{x \rightarrow 5} \frac{g(x)}{3x} = \frac{g(5)}{3 \cdot (5)} = \frac{-8}{15}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of a that contains a sum of two terms, with at least one term written as a product, and one

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term involving g and the other term involving g' (or h).

Fundamental Theorem of Calculus is evidenced by linking g' and h either directly or indirectly in the presented derivative of $g(x)$ or in $g'(5)$. Since $h(5) = -16$ is given in the question, replacing $g'(5)$ directly with -16 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(5)$ that is brought in from part (d) can earn the third point.

$25 \cdot (-16) + 10 \cdot (-8)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a , b , c , or d is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$$

$$= x^2 \cdot h(x) + 2x \cdot g(x)$$

$$a'(5) = 25 \cdot h(5) + 2 \cdot 5 \cdot g(5) = 25 \cdot (-16) + 10 \cdot (-8) = -480$$

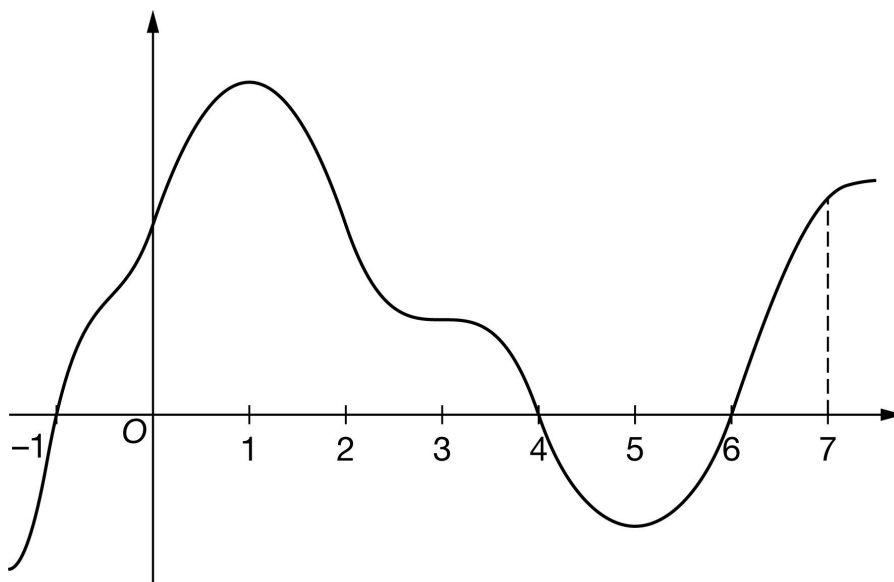
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20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-1.5 \leq x \leq 7.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-1, 4]$, $[4, 6]$, and $[6, 7]$ are 56, 10, and 8, respectively. The graph of f has horizontal tangents at $x = 1$, $x = 3$, and $x = 5$. Let h be the function defined by $h(x) = \int_6^x f(t) \, dt$ for $-1.5 \leq x \leq 7.5$.

- Find the x -coordinate of each critical point of h on the interval $-1.5 < x < 7.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of h on the interval $-1.5 < x < 7.5$. Justify your answer.
- Find the value of $h(7)$.
- Find the value of $h(-1)$. Show the computations that lead to your answer.
- Must there exist a value of d , for $-1 < d < 4$, such that $h'(d)$ is equal to the average rate of change of h over the interval $-1 \leq x \leq 4$? Justify your answer.
- Find $\lim_{x \rightarrow 7} \frac{h(x)}{4x}$. Show the computations that lead to your answer.

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(h) The function q is defined by $q(x) = 3x \cdot h(x)$. It is known that $f(7) = 14$. Find $q'(7)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1.5, 7.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1.5, 7.5)$. The point is still earned provided the three critical points, $x = -1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(4, 0)$, $(6, 0)$, read the x -coordinates and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question. m
- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum) is

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eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-1.5) > -46$
- $h(-1) > -46$
- $h(4) = 10$
- $h(6) = 0$
- $h(7) = 8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -1$, h has a relative minimum because $h' = f$ is negative for $x < -1$ and $h' = f$ is positive for $x > -1$.
- At $x = 4$, h has a relative maximum because $h' = f$ is positive for $x < 4$ and $h' = f$ is negative for $x > 4$.
- At $x = 6$, h has a relative minimum because $h' = f$ is negative for $x < 6$ and $h' = f$ is positive for $x > 6$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification

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- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 4$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

At $x = 6$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only on h' is eligible for all three points.

If a response gives justifications based only on f , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$.

Justification for the first two points should reference one of the following:

- $h'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $h'(x) = f(x)$
- h'' changes sign
- $f'(x)$ changes sign

“ $h'(x) = f(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to f as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized

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twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = 1$ and $x = 5$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = 1$ with justification
- ☐ $x = 5$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = 1$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from increasing to decreasing there.

At $x = 5$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 8 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(7) = \int_6^7 f(t) dt = 8$$

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Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 56 and ± 10 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 56) \pm (\pm 10)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 56 and ± 10 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-56 - (-10)$ earns both points because numerical expressions need not be simplified.
- The expression $(-56) - 10$ earns the first point, but not the second point.
- The expression $-56 - (-10) - 8$ earns neither point.
- The expression $\int_6^{-1} g(t) dt = -46$ earns neither point.
- The expression $-\int_{-1}^4 f(t) dt - \int_4^6 f(t) dt = -46$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 56 and ± 10 , but it does earn the second point.
- $-56, 10$ earns neither point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_4^{-1} f(t) dt \right| = 56$ and $\left| \int_4^6 f(t) dt \right| = 10$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(-1) &= \int_6^{-1} f(t) dt = - \int_{-1}^6 f(t) dt \\
 &= - \int_{-1}^4 f(t) dt - \int_4^6 f(t) dt = -56 - (-10) = -46
 \end{aligned}$$

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Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-1, 4)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(d)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(d)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = f \Rightarrow h \text{ is differentiable on } -1 \leq x \leq 4. \Rightarrow h \text{ is continuous on } -1 \leq x \leq 4.$$

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Therefore, the Mean Value Theorem can be applied to h on the interval $-1 \leq x \leq 4$ to guarantee that there exists a value of d , for $-1 < d < 4$, such that $h'(d)$ equals the average rate of change of h over the interval $-1 \leq x \leq 4$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{h(7)}{28} = \frac{2}{7}$$

$$\cdot \frac{8}{28}$$

Responses that do not earn the point include:

$$\cdot \frac{h(7)}{4 \cdot 7}$$

$$\cdot \frac{h(7)}{28}$$

Note: an incorrect value for $h(7)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = f \Rightarrow h \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 7} h(x) = h(7)$$

$$\lim_{x \rightarrow 7} \frac{h(x)}{4x} = \frac{h(7)}{4 \cdot (7)} = \frac{8}{28} = \frac{2}{7}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of q that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or f).

Fundamental Theorem of Calculus is evidenced by linking h' and f either directly or indirectly in the presented derivative of $q(x)$ or in $q'(7)$. Since $f(7) = 15$ is given in the question, replacing $h'(7)$ directly with 14 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(7)$ that is brought in from part (d) can earn the third point.

$21 \cdot 14 + 3 \cdot 8$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a , b , c , or d is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$q'(x) = 3x \cdot h'(x) + 3 \cdot h(x)$$

$$= 3x \cdot f(x) + 3 \cdot h(x)$$

$$q'(7) = 3 \cdot 7 \cdot f(7) + 3h(7) = 21 \cdot (14) + 3 \cdot (8) = 318$$

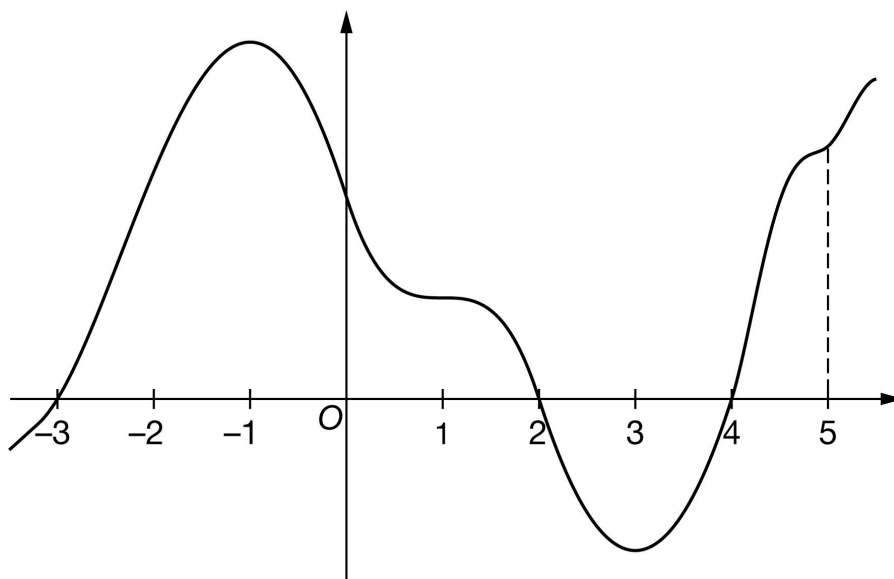
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21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 55, 12, and 10, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_4^x g(t) dt$ for $-3.5 \leq x \leq 5.5$.

- Find the x -coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.
- Find the value of $h(5)$.
- Find the value of $h(-3)$. Show the computations that lead to your answer.
- Must there exist a value of r , for $-3 < r < 2$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.
- Find $\lim_{x \rightarrow 5} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

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(h) The function k is defined by $k(x) = x \cdot h(x)$. It is known that $g(5) = 15$. Find $k'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, 5.5)$. The point is still earned provided the three critical points $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is

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eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-3.5) > -43$
- $h(-3) = -43$
- $h(2) = 12$
- $h(4) = 0$
- $h(5) = 10$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.
- At $x = 2$, h has a relative maximum because $h' = g$ h prime equals g is positive for $x < 2$ and $h' = g$ is negative for $x > 2$.
- At $x = 4$, h has a relative minimum because $h' = g$ is negative for $x < 4$ and $h' = g$ is positive for $x > 4$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

Relative minimum with justification

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Relative maximum with justification

☐

Relative minimum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 4$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

- $h'(x)$ changing from increasing to decreasing, or vice versa

- relative extrema of $h'(x) = g(x)$

- $h''(x)$ changes sign

- $g'(x)$ changes sign

“ $h'(x) = g(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to g as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn both points.

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A response that discusses h' at a particular point (i.e., $h'(-3)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 10 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

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$$h(5) = \int_4^5 g(t) dt = 10$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 55 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 55) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 55 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-55 - (-12)$ earns both points because numerical expressions need not be simplified.
- The expression $(-55) - 12$ earns the first point, but not the second point.
- The expression $-55 - (-12) - 8$ earns neither point.
- The expression $\int_4^{-3} g(t) dt = -43$ earns neither point.
- The expression $-\int_{-3}^2 g(t) dt - \int_2^4 g(t) dt = -43$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 55 and ± 12 , but it does earn the second point.
- $-55, 12$ earns neither point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_2^{-3} g(t) dt \right| = 55$ and $\left| \int_2^4 g(t) dt \right| = 12$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(-3) &= \int_4^{-3} g(t) dt = - \int_{-3}^4 g(t) dt \\
 &= - \int_{-3}^2 g(t) dt - \int_2^4 g(t) dt = -55 - (-12) = -43
 \end{aligned}$$

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Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = g \Rightarrow h \text{ is differentiable on } -3 \leq x \leq 2. \Rightarrow h \text{ is continuous on } -3 \leq x \leq 2.$$

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Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of r , for $-3 < r < 2$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{h(5)}{10} = 1$$

$$\cdot \frac{10}{10}$$

Responses that do not earn the point include:

$$\cdot \frac{h(5)}{10}$$

$$\cdot 1$$

Note: an incorrect value for $h(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h, \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 5} h(x) = h(5)$$

$$\lim_{x \rightarrow 5} \frac{h(x)}{2x} = \frac{h(5)}{2 \cdot 5} = \frac{10}{10} = 1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of k that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(5)$. Since $g(5) = 15$ is given in the question, replacing $h'(5)$ directly with 15 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(5)$ that is brought in from part (d) can earn the third point.

$5 \cdot 15 + 10$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a , b , or c is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = x \cdot h'(x) + 1 \cdot h(x)$$

$$= x \cdot g(x) + h(x)$$

$$k'(5) = 5g(5) + h(5) = 5 \cdot 15 + 10 = 85$$

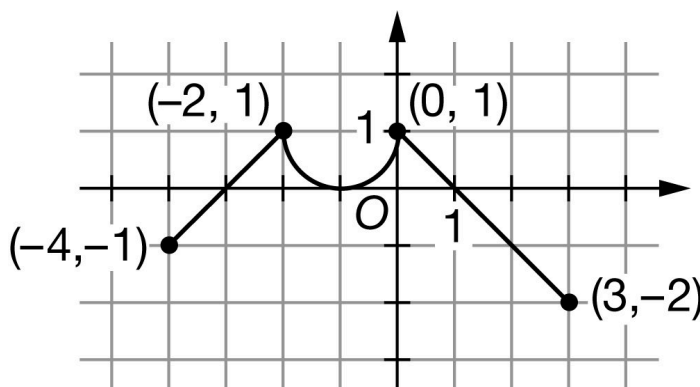
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22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f has domain $-4 \leq x \leq 3$. The graph of f , consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = -1$. Let h be

the function defined by $h(x) = \int_0^x f(t) \, dt$ for $-4 \leq x \leq 3$.

- Find the x -coordinate of each critical point of h on the interval $-4 < x < 3$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 3$, on what open intervals is h increasing and concave down? Give a reason for your answer.
- Find the value of $h(-2)$. Show the computations that lead to your answer.
- Find the value of $h(3)$. Show the computations that lead to your answer.
- Find the value of $h''(2)$, or explain why it does not exist.
- Must there exist a value of k , for $-4 < k < 0$, such that $h'(k)$ is equal to the average rate of change of h over the interval $-4 \leq x \leq 0$? Justify your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)}{3x}$. Show the computations that lead to your answer.
- The function r is defined by $r(x) = h(-3x)$. Find $r'(-1)$. Show the computations that lead to your answer.

Part A

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 3)$. The point is still earned provided the three critical points $x = -3$, $x = -1$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $f(x)$, (that is, $h' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $h' = f$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

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A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3, -1, 1$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-4) = \frac{\pi}{2} - 2$$

$$\cdot h(-3) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot h(-1) = \frac{\pi}{4} - 1$$

$$\cdot h(1) = \frac{1}{2}$$

$$\cdot h(3) = -\frac{3}{2}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.
- At $x = -1$, h has neither because $h' = f$ is positive for $x < -1$ and $h' = f$ is positive for $x > -1$.
- At $x = 1$, h has a relative maximum because $h' = f$ is positive for $x < 1$ and $h' = f$ is negative for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

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Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = -1$, h has neither because $h'(x) = f(x)$ does not change sign there.

At $x = 1$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-2, -1]$ and $(0, 1]$ or $[-2, -1)$ and $(0, 1)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

h is increasing and concave down on the open intervals $(-2, -1)$ and $(0, 1)$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(-2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(-2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(3) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\int_0^1 f(t) dt = \frac{1}{2}$ – OR – $\int_1^3 f(t) dt = -2$

☐ Answer

Solution:

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$$h(3) = \int_0^3 f(t) dt$$

$$= \int_0^1 f(t) dt + \int_1^3 f(t) dt = \frac{1}{2} + (-2) = -\frac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(2) = -1$ or $f'(2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h''(x) = f'(x)$$

$$h''(2) = f'(2) = \frac{1 - (-2)}{0 - 3} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$ ” earns both points.

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- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = f \Rightarrow h \text{ is differentiable on } -4 \leq x \leq 0. \Rightarrow h \text{ is continuous on } -4 \leq x \leq 0.$$

Therefore, the Mean Value Theorem can be applied to h on the interval $-4 \leq x \leq 0$ to guarantee that there exists a value of k , for $-4 < k < 0$, such that $h'(k)$ equals the average rate of change of h over the interval $-4 \leq x \leq 0$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow -2} \frac{h'(x)}{3x} = \frac{-1}{6}$ earns the point, as does just stating the answer of $\frac{-1}{6}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

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0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = f \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$$

$$\lim_{x \rightarrow -2} \frac{h'(x)}{3x} = \frac{h'(-2)}{3 \cdot (-2)} = \frac{f(-2)}{-6} = \frac{1}{-6}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(-3x) \cdot (-3)$$

$$\cdot f(-3x) \cdot (-3)$$

$$\cdot h'(-3 \cdot -1) \cdot (-3)$$

$$\cdot f(-3 \cdot -1) \cdot (-3)$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot (-3)$$

$$\cdot f(x) \cdot (-3)$$

$$\cdot h'(3) \cdot (-3)$$

$$\cdot f(3) \cdot (-3)$$

$$\cdot h(-3x) \cdot (-3)$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(-1)$. For example, the second point is earned by $f(-3x)$ or by $f(-3 \cdot -1)$.

The second point may be earned without earning the first point. For example, $h'(-3x) = f(-3x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are

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linked.

The second point may be earned by going directly from $h'(3)$ to -2 . The function f does not have to be stated explicitly. For example, $h'(-3 \cdot -1) \cdot (-3) = -2 \cdot (-3)$ earns the first two points. Recall, if $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $f(-3x) \cdot (-3)$ and $f(-3 \cdot -1) \cdot (-3)$ earn the first two points.

The third point is earned for the correct answer of 6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$r'(x) = h'(-3x) \cdot (-3)$$

$$= f(-3x) \cdot (-3)$$

$$r'(-1) = f((-3) \cdot (-1)) \cdot (-3) = f(3) \cdot (-3) = -2 \cdot (-3) = 6$$

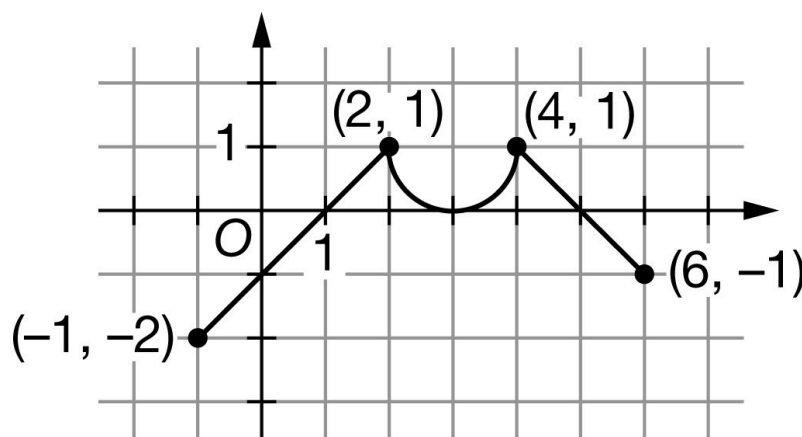
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23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The continuous function h has domain $-1 \leq x \leq 6$. The graph of h , consisting of two line segments and a semicircle, is shown in the figure above. The graph of h has a horizontal tangent at $x = 3$. Let g be the function defined by $g(x) = \int_2^x h(t) \, dt$ for $-1 \leq x \leq 6$.

- Find the x -coordinate of each critical point of g on the interval $-1 \leq x \leq 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-1 \leq x \leq 6$, on what open intervals is g decreasing and concave up? Give a reason for your answer.
- Find the value of $g(4)$. Show the computations that lead to your answer.
- Find the value of $g(-1)$. Show the computations that lead to your answer.
- Find the value of $g''(0)$, or explain why it does not exist.
- Must there exist a value of d , for $-1 < d < 3$, such that $g'(d)$ is equal to the average rate of change of g over the interval $-1 \leq x \leq 3$? Justify your answer.
- Find $\lim_{x \rightarrow 4} \frac{g'(x)}{3x}$. Show the computations that lead to your answer.
- The function a is defined by $a(x) = g(2x)$. Find $a'(0)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 3$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 6)$. The point is still earned provided the three critical points $x = 1$, $x = 3$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(3, 0)$, and $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = 1$, $x = 3$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $h(x)$, (that is, $g' = h$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $g' = h$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 1$, g has a relative minimum) is eligible for at most one of the three points.

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Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ value for $x = 1, 3, 5$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot g(-1) = \frac{3}{2}$$

$$\cdot g(1) = -\frac{1}{2}$$

$$\cdot g(3) = 1 - \frac{\pi}{4}$$

$$\cdot g(5) = \frac{5}{2} - \frac{\pi}{4}$$

$$\cdot g(6) = 2 - \frac{\pi}{4}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = 1$, g has a relative minimum because $g' = h$ is negative for $x < 1$ and $g' = h$ is positive for $x > 1$.
- At $x = 3$, g has neither because $g' = h$ is positive for $x < 3$ and $g' = h$ is positive for $x > 3$.
- At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification

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- ☐ Relative maximum with justification

Solution:

At $x = 1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 3$, g has neither because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 1]$, $(-1, 1]$, $[-1, 1)$, and $(-1, 1)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only h is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ g' is increasing” can also be expressed by either “ g'' is positive” or “the slope of g' is positive.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where $g' = h$ is negative.

g is concave up where $g' = h$ is increasing.

g is decreasing and concave up on the open interval $(-1, 1)$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g(4) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(4) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-1) = \frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_{-1}^1 h(t) dt = -2$ – OR – $\int_1^2 h(t) dt = \frac{1}{2}$

☐ Answer

Solution:

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$$\begin{aligned}
 g(-1) &= \int_2^{-1} h(t) \, dt = - \int_{-1}^2 h(t) \, dt \\
 &= - \int_{-1}^1 h(t) \, dt - \int_1^2 h(t) \, dt = -(-2) - \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(0) = 1$ or $h'(0) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g''(x) = h'(x)$$

$$g''(0) = h'(0) = \frac{1}{2} = \frac{1 - (-2)}{2 - (-1)}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” earns both points.

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- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ g is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow g$ is continuous on $-1 \leq x \leq 3$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-1 \leq x \leq 3$ to guarantee that there exists a value of d , for $-1 < d < 3$, such that $g'(d)$ equals the average rate of change of g over the interval $-1 \leq x \leq 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 4} \frac{g'(x)}{3x} = \frac{1}{12}$ earns the point, as does just stating the answer of $\frac{1}{12}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.

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0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g' = h \Rightarrow g' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 4} g'(x) = g'(4)$$

$$\lim_{x \rightarrow 4} \frac{g'(x)}{3x} = \frac{g'(4)}{3 \cdot 4} = \frac{h(4)}{12} = \frac{1}{12}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 0$ is substituted for x , then $x = 0$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot g'(2x) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

$$\cdot g'(2 \cdot 0) \cdot 2$$

$$\cdot h(2 \cdot 0) \cdot 2$$

Responses that do not earn the point include:

$$\cdot g'(x) \cdot 2$$

$$\cdot h(x) \cdot 2$$

$$\cdot g'(0) \cdot 2$$

$$\cdot h(0) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and h either directly or indirectly in the presented derivative of $a(x)$ or in $a'(0)$. For example, the second point is earned by $h(2x)$ or by $h(2 \cdot 0)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

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The second point may be earned by going directly from $g'(0)$ to -1 . The function h does not have to be stated explicitly. For example, $g'(2 \cdot 0) \cdot 2 = -1 \cdot 2$ earns the first two points. Recall, if $x = 0$ is substituted for x , then $x = 0$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 0) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -2 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = g'(2x) \cdot 2$$

$$= h(2x) \cdot 2$$

$$a'(0) = g'(2 \cdot 0) \cdot 2 = g'(0) \cdot 2 = h(0) \cdot 2 = -1 \cdot 2 = -2$$

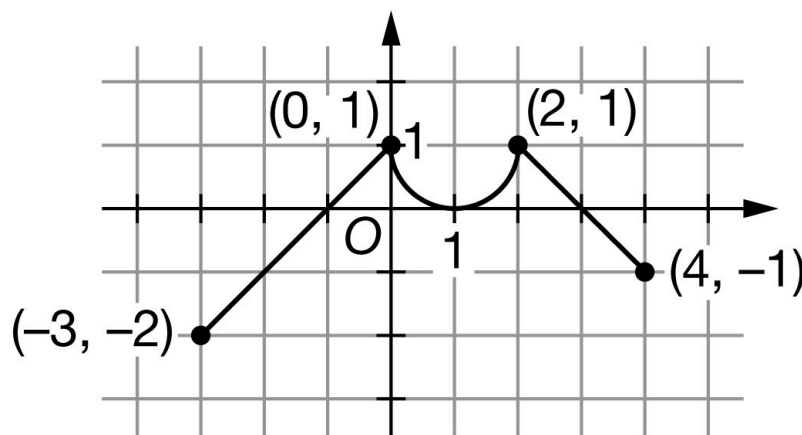
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24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The continuous function g has domain $-3 \leq x \leq 4$. The graph of g , consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = 1$. Let h be the function defined by $h(x) = \int_0^x g(t) \, dt$ for $-3 \leq x \leq 4$.

- Find the x -coordinate of each critical point of h on the interval $-3 < x < 4$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-3 < x < 4$, on what open intervals is h decreasing and concave down? Give a reason for your answer.
- Find the value of $h(2)$. Show the computations that lead to your answer.
- Find the value of $h(-3)$. Show the computations that lead to your answer.
- Find the value of $h''(-1)$, or explain why it does not exist.
- Must there exist a value of r , for $-3 < r < 1$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow 2} \frac{h'(x)}{5x}$. Show the computations that lead to your answer.
- The function k is defined by $k(x) = h(x^2)$. Find $k'(-2)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -1$, $x = 1$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum) is eligible for at most one of the three points.

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Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -1, 1, 3$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-3) = \frac{3}{2}$$

$$\cdot h(-1) = -\frac{1}{2}$$

$$\cdot h(1) = 1 - \frac{\pi}{4}$$

$$\cdot h(3) = \frac{5}{2} - \frac{\pi}{2}$$

$$\cdot h(4) = 2 - \frac{\pi}{2}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -1$, h has a relative minimum because $h' = g$ is negative for $x < -1$ and $h' = g$ is positive for $x > -1$.
- At $x = 1$, h has neither because $h' = g$ is positive for $x < 1$ and $h' = g$ is positive for $x > 1$.
- At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification

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☐ Relative maximum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[3, 4]$, $(3, 4]$, $[3, 4)$, and $(3, 4)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is decreasing where $h' = g$ is negative.

h is concave down where $h' = g$ is decreasing.

h is decreasing and concave down on the open interval $(3, 4)$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(2) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(2) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-3) = \frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\int_{-3}^{-1} g(t) dt = -2$ – OR – $\int_{-1}^0 g(t) dt = \frac{1}{2}$

☐ Answer

Solution:

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$$\begin{aligned}
 h(-3) &= \int_0^{-3} g(t) \, dt = - \int_{-3}^0 g(t) \, dt \\
 &= - \int_{-3}^{-1} g(t) \, dt - \int_{-1}^0 g(t) \, dt = -(-2) - \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-1) = 1$ or $g'(-1) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-1) = g'(-1) = \frac{1 - (-2)}{0 - (-3)} = 1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h'(x)$ ” earns both points.

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- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1 \Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of r , for $-3 < r < 1$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{1}{10}$ earns the point, as does just stating the answer of $\frac{1}{10}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

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0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 2} h'(x) = h'(2)$$

$$\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{h'(2)}{5 \cdot 2} = \frac{g(2)}{10} = \frac{1}{10}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor in order to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot g((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot h'(4) \cdot 2 \cdot (-2)$$

$$\cdot g(4) \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4)$$

$$\cdot g((-2)^2) \cdot (-4)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

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$$\cdot g(x) \cdot 2x$$

$$\cdot h'(4) \cdot (-4)$$

$$\cdot g(4) \cdot (-4)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-2)$. For example, the second point is earned by $g(x^2)$ or by $g((-2)^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = g(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and Ag are linked.

The second point may be earned by going directly from $h'(4)$ to -1 . The function g does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2) = -1 \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4) = -1 \cdot (-4)$$

Recall, if $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g(x^2) \cdot 2x$ and $g((-2)^2) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 4 simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Unit 1 AP Topics 5-8**Solution:**

$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-2) = g((-2)^2) \cdot 2 \cdot (-2) = g(4) \cdot (-4) = -1 \cdot (-4) = 4$$

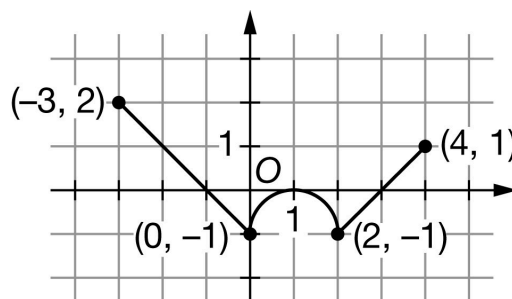
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25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f has domain $-3 \leq x \leq 4$. The graph of f , consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = 1$. Let g be the function defined by $g(x) = \int_0^x f(t) \, dt$ for $-3 \leq x \leq 4$.

- Find the x -coordinate of each critical point of g on the interval $-3 < x < 4$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-3 < x < 4$, on what open intervals is g decreasing and concave up? Give a reason for your answer.
- Find the value of $g(2)$. Show the computations that lead to your answer.
- Find the value of $g(-3)$. Show the computations that lead to your answer.
- Find the value of $g''(-2)$, or explain why it does not exist.
- Must there exist a value of b , for $-3 < b < 1$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow 2} \frac{g'(x)}{4x}$. Show the computations that lead to your answer.
- The function z is defined by $z(x) = g(x^2)$. Find $z'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -1$, $x = 1$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $f(x)$, (that is, $g' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $g' = f$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, f has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

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A response that discusses g' at a particular point (e.g., “ $g'(3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ values for $x = -1, 1, 3$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot g(-3) = -\frac{3}{2}$$

$$\cdot g(-1) = \frac{1}{2}$$

$$\cdot g(1) = \frac{\pi}{4} - 1$$

$$\cdot g(3) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot g(4) = \frac{\pi}{2} - 2$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -1$, g has a relative maximum because $g' = f$ is positive for $x < -1$ and $g' = f$ is negative for $x > -1$.
- At $x = 1$, g has neither because $g' = f$ is negative for $x < 1$ and $g' = f$ is negative for $x > 1$.
- At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

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At $x = 1$, g has neither because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[0, 1]$ and $(2, 3]$ or $[0, 1]$ and $(2, 3)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

The reason “ g' is increasing” can also be expressed by either “ g'' is positive” or “the slope of g' is positive.”

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

g is decreasing where $g' = f$ is negative.

g is concave up where $g' = f$ is increasing.

g is decreasing and concave up on the open intervals $(0, 1)$ and $(2, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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Correct supporting work is needed to earn this point.

The correct decimal value is $g(2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

$$g(2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, 2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-3) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_{-3}^{-1} f(t) dt = 2$ – OR – $\int_{-1}^0 f(t) dt = -\frac{1}{2}$

☐ Answer

Solution:

$$\begin{aligned} g(-3) &= \int_0^{-3} f(t) dt = -\int_{-3}^0 f(t) dt \\ &= -\int_{-3}^{-1} f(t) dt - \int_{-1}^0 f(t) dt = -2 - \left(-\frac{1}{2}\right) = -\frac{3}{2} \end{aligned}$$

Part F

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(-2) = -1$ or $f'(-2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''(-2) = f'(-2) = \frac{-1-2}{0-(-3)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.

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· “Yes, because $g(x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ g is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow g$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of b , for $-3 < b < 1$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{g'(x)}{4x} = \frac{-1}{8}$ earns the point, as does just stating the answer of $\frac{-1}{8}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

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$$g' = f \Rightarrow g' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 2} g'(x) = g'(2)$$

$$\lim_{x \rightarrow 2} \frac{g'(x)}{4x} = \frac{g'(2)}{4 \cdot 2} = \frac{f(2)}{8} = \frac{-1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

- $g'(x^2) \cdot 2x$
- $f(x^2) \cdot 2x$
- $g'((-2)^2) \cdot 2 \cdot (-2)$
- $f((-2)^2) \cdot 2 \cdot (-2)$
- $g'(4) \cdot 2 \cdot (-2)$
- $f(4) \cdot 2 \cdot (-2)$
- $g'((-2)^2) \cdot (-4)$
- $f((-2)^2) \cdot (-4)$

Responses that do not earn the point include:

- $g'(x^2) \cdot 2$
- $f(x^2) \cdot 2$
- $g'(x) \cdot 2x$
- $f(x) \cdot 2x$
- $g'(4) \cdot (-4)$
- $f(4) \cdot (-4)$
- $g(x^2) \cdot 2x$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $z(x)$ or in $z'(-2)$. For example, the second point is earned by $f(x^2)$ or by $f((-2)^2)$.

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The second point may be earned without earning the first point. For example, $g'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(4)$ to 1. The function f does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot g'((-2)^2) \cdot 2 \cdot (-2) = 1 \cdot 2 \cdot (-2)$$

$$\cdot g'((-2)^2) \cdot (-4) = 1 \cdot (-4)$$

Recall, if $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f((-2)^2) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of -4 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $g'(x)^2 \cdot 2x$ or $(g'(x))^2 \cdot 2x$ are not uncommon. These type of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$z'(x) = g'(x^2) \cdot 2x$$

$$= f(x^2) \cdot 2x$$

$$z'(-2) = f((-2)^2) \cdot 2 \cdot (-2) = f(4) \cdot (-4) = 1 \cdot (-4) = -4$$

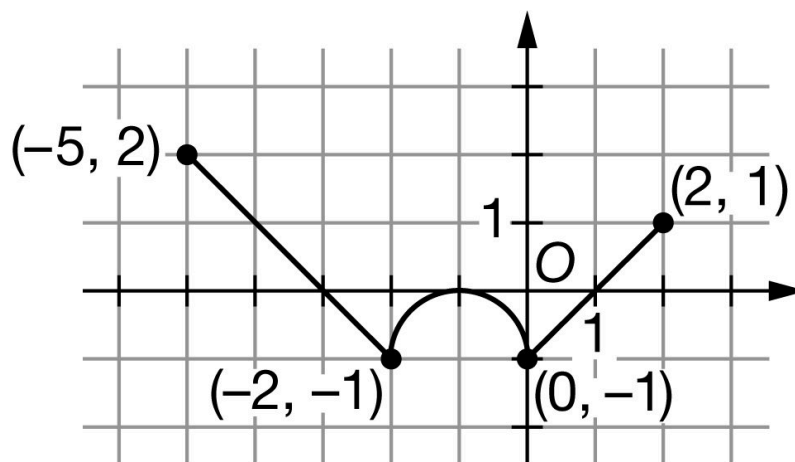
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26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The continuous function g has domain $-5 \leq x \leq 2$. The graph of g , consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = -1$. Let h be the function defined by $h(x) = \int_{-2}^x g(t) \, dt$ for $-5 \leq x \leq 2$.

- Find the x -coordinate of each critical point of h on the interval $-5 < x < 2$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-5 < x < 2$, on what open intervals is h increasing and concave up? Give a reason for your answer.
- Find the value of $h(0)$. Show the computations that lead to your answer.
- Find the value of $h(-5)$. Show the computations that lead to your answer.
- Find the value of $h''(-3)$, or explain why it does not exist.
- Must there exist a value of p , for $-3 < p < 1$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)}{4x}$. Show the computations that lead to your answer.
- The function f is defined by $f(x) = h(-2x)$. Find $f'(1)$. Show the computations that lead to your answer.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 2)$. The point is still earned provided the three critical points $x = -3$, $x = -1$ and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$ and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

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Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3, -1, 1$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-5) = -\frac{3}{2}$$

$$\cdot h(-3) = \frac{1}{2}$$

$$\cdot h(-1) = \frac{\pi}{4} - 1$$

$$\cdot h(1) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot h(2) = \frac{\pi}{2} - 2$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -3$, h has a relative maximum because $h' = g$ is positive for $x < -3$ and $h' = g$ is negative for $x > -3$.
- At $x = -1$, h has neither because $h' = g$ is negative for $x < -1$ and $h' = g$ is negative for $x > -1$.
- At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification

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- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $(1, 2]$, $[1, 2)$, and $(1, 2)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is increasing” can also be expressed by either “ h'' is positive” or “the slope of h' is positive.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave up where $h' = g$ is increasing.

h is increasing and concave up on the open interval $(1, 2)$.

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Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(0) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(0) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-5) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\int_{-5}^{-3} g(t) dt = 2$ – OR – $\int_{-3}^{-2} g(t) dt = -\frac{1}{2}$

☐ Answer

Solution:

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$$\begin{aligned}
 h(-5) &= \int_{-2}^{-5} g(t) \, dt = - \int_{-5}^{-2} g(t) \, dt \\
 &= - \int_{-5}^{-3} g(t) \, dt - \int_{-3}^{-2} g(t) \, dt = -(2) - \left(-\frac{1}{2}\right) = -\frac{3}{2}
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-3) = -1$ or $g'(-3) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-3) = g'(-3) = \frac{2 - (-1)}{-5 - (-2)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” earns both points.

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- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of p , for $-3 < p < 1$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{h'(x)}{4x} = \frac{1}{8}$ earns the point, as does just stating the answer of $\frac{1}{8}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

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0

1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$$

$$\lim_{x \rightarrow -2} \frac{h'(x)}{4x} = \frac{h'(-2)}{4 \cdot (-2)} = \frac{g(-2)}{-8} = \frac{-1}{-8} = \frac{1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(-2x) \cdot (-2)$$

$$\cdot g(-2x) \cdot (-2)$$

$$\cdot h'(-2 \cdot 1) \cdot (-2)$$

$$\cdot g(-2 \cdot 1) \cdot (-2)$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot (-2)$$

$$\cdot g(x) \cdot (-2)$$

$$\cdot h'(-2) \cdot (-2)$$

$$\cdot g(-2) \cdot (-2)$$

$$\cdot h(-2x) \cdot (-2)$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(1)$. For example, the second point is earned by $g(-2x)$ or by $g(-2 \cdot 1)$.

The second point may be earned without earning the first point. For example, $h'(-2x) = g(-2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are

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linked.

The second point may be earned by going directly from $h'(-2)$ to -1 . The function g does not have to be stated explicitly. For example, $h'(-2 \cdot 1) \cdot (-2) = -1 \cdot (-2)$ earns the first two points. Recall, if $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $g(-2x) \cdot (-2)$ and $g(-2 \cdot 1) \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = h'(-2x) \cdot (-2)$$

$$= g(-2x) \cdot (-2)$$

$$f'(1) = g(-2 \cdot 1) \cdot (-2) = g(-2) \cdot (-2) = (-1) \cdot (-2) = 2$$

27. Let g and h be the functions defined by $g(x) = \sin\left(\frac{\pi}{2}x\right) + 4$ and $h(x) = -\frac{1}{4}x^3 + \frac{3}{4}x + \frac{9}{2}$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-1 < x < 2$, what is $\lim_{x \rightarrow 1} f(x)$?

- (A) 4
- (B) $\frac{9}{2}$

(C) 5



- (D) The limit cannot be determined from the information given.

Answer C

Correct. Since $\lim_{x \rightarrow 1} g(x) = \sin\left(\frac{\pi}{2}\right) + 4 = 1 + 4 = 5$ and $\lim_{x \rightarrow 1} h(x) = -\frac{1}{4} + \frac{3}{4} + \frac{9}{2} = 5$, it can be concluded from the squeeze theorem that $\lim_{x \rightarrow 1} f(x) = 5$ because $g(x) \leq f(x) \leq h(x)$ for $-1 < x < 2$.

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28. The function f is defined for all x in the interval $3 < x < 6$. Which of the following statements, if true, implies that $\lim_{x \rightarrow 5} f(x) = 12$?
- (A) There exists a function g with $f(x) \leq g(x)$ for $3 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = 12$.
- (B) There exists a function g with $g(x) \leq f(x)$ for $3 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = 12$.
- (C) There exist functions g and h with $g(x) \leq f(x) \leq h(x)$ for $3 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = 11$ and $\lim_{x \rightarrow 5} h(x) = 13$.
- (D) There exist functions g and h with $g(x) \leq f(x) \leq h(x)$ for $3 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = \lim_{x \rightarrow 5} h(x) = 12$. ✓

Answer D

Correct. The hypothesis conditions of the squeeze theorem are confirmed, so it may be concluded that $\lim_{x \rightarrow 5} f(x) = 12$.

29. The function f is defined for all x in the interval $4 < x < 6$. Which of the following statements, if true, implies that $\lim_{x \rightarrow 5} f(x) = 17$?
- (A) There exists a function g with $f(x) \leq g(x)$ for $4 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = 17$.
- (B) There exists a function g with $g(x) \leq f(x)$ for $4 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = 17$.
- (C) There exist functions g and h with $f(x) \leq g(x) \leq h(x)$ for $4 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = \lim_{x \rightarrow 5} h(x) = 17$.
- (D) There exist functions g and h with $g(x) \leq f(x) \leq h(x)$ for $4 < x < 6$, and $\lim_{x \rightarrow 5} g(x) = \lim_{x \rightarrow 5} h(x) = 17$. ✓

Answer D

Correct. The hypothesis conditions of the squeeze theorem are confirmed so it may be concluded that $\lim_{x \rightarrow 5} f(x) = 17$.

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30. The function g is given by $g(x) = \frac{7x-26}{x-5}$. The function h is given by $h(x) = \frac{3x+14}{2x+1}$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $0 < x < 5$, what is $\lim_{x \rightarrow 2} f(x)$?

(A) $\frac{3}{2}$

(B) 4

(C) 7

(D) The limit cannot be determined from the information given.

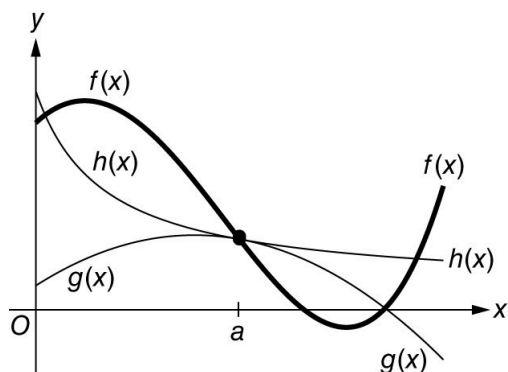
Answer B

Correct. Since $\lim_{x \rightarrow 2} g(x) = \frac{14-26}{2-5} = \frac{-12}{-3} = 4$ and $\lim_{x \rightarrow 2} h(x) = \frac{6+14}{4+1} = \frac{20}{5} = 4$, it can be concluded from the squeeze theorem that $\lim_{x \rightarrow 2} f(x) = 4$ because $g(x) \leq f(x) \leq h(x)$ for $0 < x < 5$.

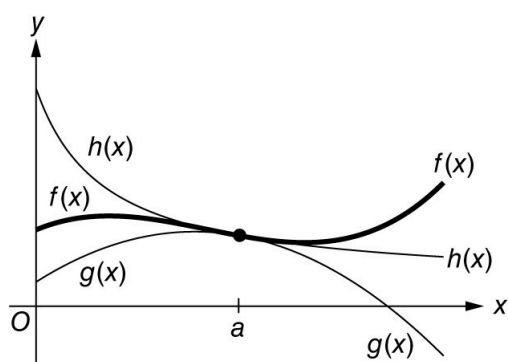
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31. Let f be a function of x . The value of $\lim_{x \rightarrow a} f(x)$ can be found using the squeeze theorem with the functions g and h . Which of the following could be graphs of f , g , and h ?

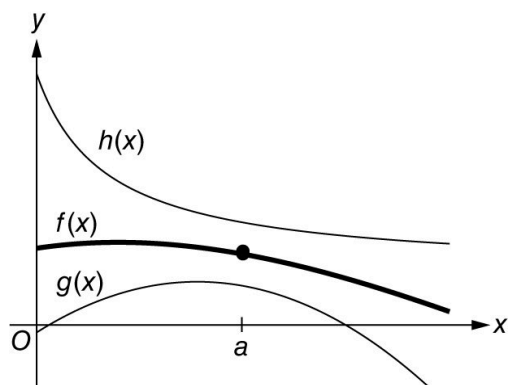
(A)



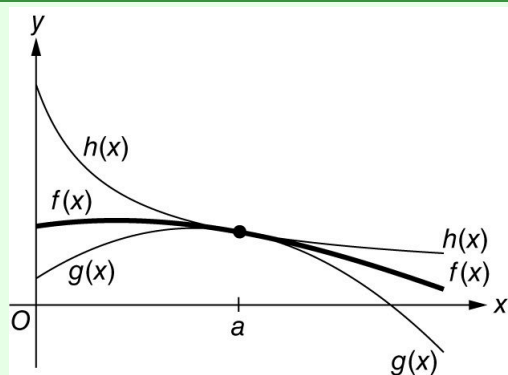
(B)



(C)



(D)



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Answer D

Correct. The hypothesis conditions of the squeeze theorem are satisfied, since $g(x) \leq f(x) \leq h(x)$ on the interval shown and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x)$. Therefore $\lim_{x \rightarrow a} f(x)$ can be found using the squeeze theorem with these functions g and h .

32. Let f , g , and h be the functions defined by $f(x) = \frac{1 - \cos x}{x^2}$, $g(x) = x^2 \sin\left(\frac{1}{x}\right)$, and $h(x) = \frac{\sin x}{x}$ for $x \neq 0$. All of the following inequalities are true for $x \neq 0$. Which of the inequalities can be used with the squeeze theorem to find the limit of the function as x approaches 0?

- I. $\frac{1}{3}(1 - x^2) \leq f(x) \leq \frac{1}{2}$
II. $-x^2 \leq g(x) \leq x^2$
III. $-\frac{1}{|x|} \leq h(x) \leq \frac{1}{|x|}$

(A) I only

(B) II only

(C) I and III only

(D) II and III only



Answer B

Correct. Because $\lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} (x^2) = 0$, it can be concluded from the squeeze theorem that $\lim_{x \rightarrow 0} g(x) = 0$.

Because $\lim_{x \rightarrow 0} \frac{1}{3}(1 - x^2) = \frac{1}{3} \neq \frac{1}{2}$, the conditions of the squeeze theorem are not satisfied for inequality (I) and the theorem cannot be used to find $\lim_{x \rightarrow 0} f(x)$.

Because $\lim_{x \rightarrow 0} \frac{1}{|x|}$ does not exist, the conditions of the squeeze theorem are not satisfied for inequality (III) and the theorem cannot be used to find $\lim_{x \rightarrow 0} h(x)$.

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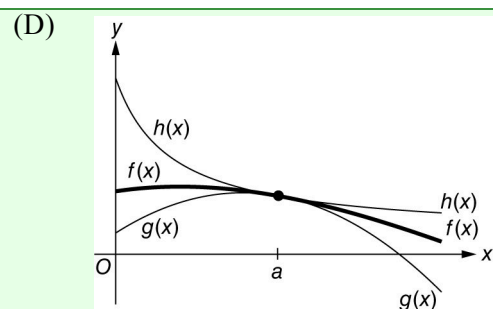
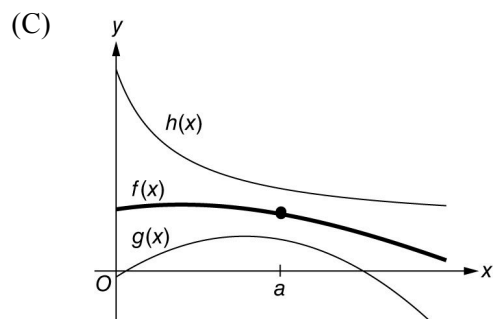
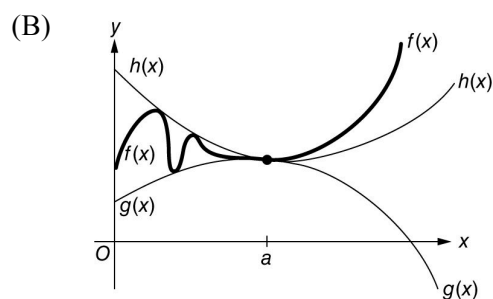
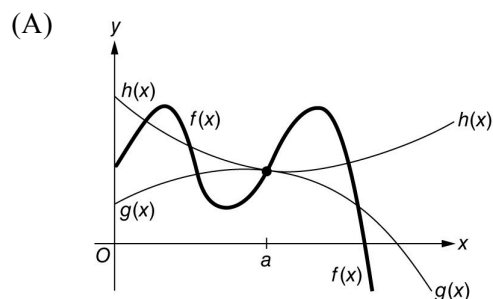
33. Let g and h be the functions defined by $g(x) = -x^2 - 2x + 3$ and $h(x) = \frac{1}{2}x^2 + x + \frac{13}{2}$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow -1} f(x)$?
- (A) 4
- (B) 5
- (C) 6
- (D) The limit cannot be determined from the information given. ✓

Answer D

Correct. Because $\lim_{x \rightarrow -1} g(x) = -(-1)^2 - 2(-1) + 3 = 4$ and $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}(-1)^2 + (-1) + \frac{13}{2} = 6$, the conditions for the squeeze theorem are not satisfied and no conclusion can be made about the exact value of $\lim_{x \rightarrow -1} f(x)$.

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34. Let f be a function of x . The value of $\lim_{x \rightarrow a} f(x)$ can be found using the squeeze theorem with the functions g and h . Which of the following could be graphs of f , g , and h ?

**Answer D**

Correct. The hypothesis conditions of the squeeze theorem are satisfied, since $g(x) \leq f(x) \leq h(x)$ on the interval shown and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x)$. Therefore, $\lim_{x \rightarrow a} f(x)$ can be found using the squeeze theorem with these functions g and h .

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35. The function g is given by $g(x) = \frac{1}{x^2 - 4x + 5}$. The function h is given by $h(x) = \frac{2x^2 - 8x + 10}{x^2 - 4x + 6}$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $0 < x < 5$, what is $\lim_{x \rightarrow 2} f(x)$?
- (A) 0
- (B) 1
- (C) 2
- (D) The limit cannot be determined from the information given.

Answer B

Correct. Since $\lim_{x \rightarrow 2} g(x) = \frac{1}{2^2 - (4)(2) + 5} = \frac{1}{1} = 1$ and $\lim_{x \rightarrow 2} h(x) = \frac{2(2^2) - (8)(2) + 10}{2^2 - (4)(2) + 6} = \frac{2}{2} = 1$, it can be concluded from the squeeze theorem that $\lim_{x \rightarrow 2} f(x) = 1$ because $g(x) \leq f(x) \leq h(x)$ for $0 < x < 5$.

36. Let g and h be the functions defined by $g(x) = -2x^2 + 4x + 1$ and $h(x) = \frac{1}{2}x^2 - x + \frac{11}{2}$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow 1} f(x)$?
- (A) 3
- (B) 4
- (C) 5
- (D) The limit cannot be determined from the information given.

Answer D

Correct. Because $\lim_{x \rightarrow 1} g(x) = -2(1) + 4(1) + 1 = 3$ and $\lim_{x \rightarrow 1} h(x) = \frac{1}{2}(1) - 1 + \frac{11}{2} = \frac{10}{2} = 5$, the conditions for the squeeze theorem are not satisfied and no conclusion can be made about the exact value of $\lim_{x \rightarrow 1} f(x)$.

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37. Let f , g , and h be the functions defined by $f(x) = \frac{\sin x}{2x}$, $g(x) = x^4 \cos\left(\frac{1}{x^2}\right)$, and $h(x) = \frac{x^2}{\tan x}$ for $x \neq 0$. All of the following inequalities are true on the interval $[-1, 1]$ for $x \neq 0$. Which of the inequalities can be used with the squeeze theorem to find the limit of the function as x approaches 0?

- I. $\frac{1}{4} \leq f(x) \leq x^2 + \frac{1}{2}$
 II. $-x^4 \leq g(x) \leq x^4$
 III. $-\frac{1}{x^2} \leq h(x) \leq \frac{1}{x^2}$

(A) I only

(B) II only

(C) I and III only

(D) II and III only



Answer B

Correct. Because $\lim_{x \rightarrow 0} (-x^4) = 0 = \lim_{x \rightarrow 0} x^4$, it can be concluded from the squeeze theorem that $\lim_{x \rightarrow 0} g(x) = 0$.

Because $\lim_{x \rightarrow 0} x^2 + \frac{1}{2} = \frac{1}{2} \neq \frac{1}{4}$, the conditions of the squeeze theorem are not satisfied for inequality (I) and the theorem cannot be used to find $\lim_{x \rightarrow 0} f(x)$. Because $\lim_{x \rightarrow 0} -\frac{1}{x^2}$ and $\lim_{x \rightarrow 0} \frac{1}{x^2}$ do not exist, the conditions of the squeeze theorem are not satisfied for inequality (III) and the theorem cannot be used to find $\lim_{x \rightarrow 0} h(x)$.

38. Let g and h be the functions defined by $g(x) = \sin\left(\frac{\pi}{2}(x+2)\right) + 3$ and $h(x) = -\frac{1}{4}x^3 - \frac{3}{2}x^2 - \frac{9}{4}x + 3$. If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-2 < x < 0$, what is $\lim_{x \rightarrow -1} f(x)$?

(A) 3

(B) 3.5

(C) 4

(D) The limit cannot be determined from the information given.



Answer C

Correct. Since $\lim_{x \rightarrow -1} g(x) = \sin\left(\frac{\pi}{2}(-1+2)\right) + 3 = \sin\left(\frac{\pi}{2}\right) + 3 = 1 + 3 = 4$ and

$\lim_{x \rightarrow -1} h(x) = \frac{1}{4} - \frac{3}{2} + \frac{9}{4} + 3 = \frac{1-6+9+12}{4} = \frac{16}{4} = 4$, it can be concluded from the squeeze theorem that

$\lim_{x \rightarrow -1} f(x) = 4$ because $g(x) \leq f(x) \leq h(x)$ for $-2 < x < 0$.

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39. $\lim_{x \rightarrow 0} \frac{2x^6 + 6x^3}{4x^5 + 3x^3}$ is

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

(E) nonexistent



40. $\lim_{x \rightarrow 0} \frac{5x^4 + 8x^2}{3x^4 - 16x^2}$ is

(A) $-\frac{1}{2}$

(B) 0

(C) 1

(D) $\frac{5}{3}$

(E) nonexistent



41. If $f(x) = \begin{cases} \ln x & \text{for } 0 < x \leq 2 \\ x^2 \ln 2 & \text{for } 2 < x \leq 4, \end{cases}$ then $\lim_{x \rightarrow 2} f(x)$ is

(A) $\ln 2$

(B) $\ln 8$

(C) $\ln 16$

(D) 4

(E) nonexistent



Unit 1 AP Topics 5-8

42.

$\lim_{x \rightarrow -5} f(x) = 4$	$\lim_{x \rightarrow 5} f(x) = 2$	$\lim_{x \rightarrow 5} g(x) = 5$
------------------------------------	-----------------------------------	-----------------------------------

The table above gives selected limits of the functions f and g . What is $\lim_{x \rightarrow 5} (f(-x) + 3g(x))$?

(A) 19

(B) 17

(C) 13

(D) 9

Answer A

This option is correct. Calculating this limit uses the basic properties for the limit of the sum of two functions, the limit of a product of a function and a constant, and the limit of the reflection of a function across the y -axis:

$$\lim_{x \rightarrow 5} (f(-x) + 3g(x)) = \lim_{x \rightarrow 5} f(-x) + \lim_{x \rightarrow 5} (3g(x))$$

$$= \lim_{x \rightarrow -5} f(x) + 3 \lim_{x \rightarrow 5} g(x) = 4 + 3(5) = 19.$$

43.

$$\lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3} \text{ is}$$

(A) -3 (B) -1 (C) 1 (D) 3

(E) nonexistent

44. If f is a continuous function such that $f(2) = 6$, which of the following statements must be true?

(A) $\lim_{x \rightarrow 1} f(2x) = 3$

(B) $\lim_{x \rightarrow 2} f(2x) = 12$

(C) $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = 6$

(D) $\lim_{x \rightarrow 2} f(x^2) = 36$

(E) $\lim_{x \rightarrow 2} (f(x))^2 = 36$

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45. If $a \neq 0$, then $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x^4 - a^4}$ is

(A) $\frac{1}{a^2}$

(B) $\frac{1}{2a^2}$ ✓

(C) $\frac{1}{6a^2}$

(D) 0

(E) nonexistent

46. $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4}$ is

(A) $-\frac{1}{4}$

(B) 0

(C) 1

(D) $\frac{5}{4}$ ✓

(E) nonexistent

47. $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x^2 - 2x - 15}$ is

(A) 0

(B) $\frac{3}{5}$

(C) $\frac{3}{4}$ ✓

(D) 1

(E) nonexistent

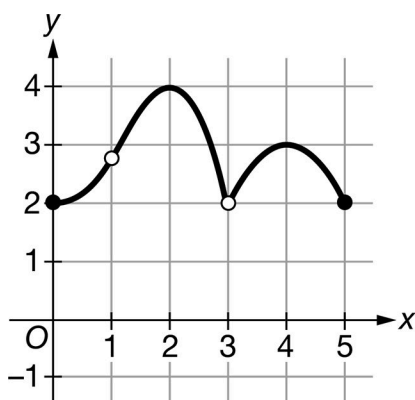
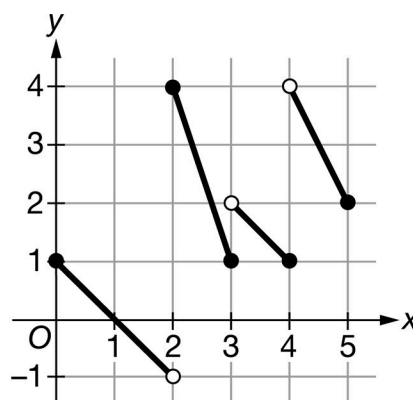
Unit 1 AP Topics 5-8

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f Graph of g

The graphs of the functions f and g are shown above on the interval $0 \leq x \leq 5$.

- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 2.5$.
- Let h be the function defined by $h(x) = g(f(x))$. Find the value of $\lim_{x \rightarrow 3} h(x)$ or explain why the limit does not exist. Use correct limit notation in your answer.
- Let k be the function defined by $k(x) = (4 - f(x))g(x)$. Consider $x = 2$ and $x = 4$. Determine whether k is continuous at each of these values. Justify your answers using correct limit notation.
- Values of $f(x)$ at selected values of x are given in the table below. What type of discontinuity does f have at $x = 1$? Based on the values in the table, what is a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$? Give a reason for your answer.

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$	2.634	2.770	2.784	Undefined	2.787	2.801	2.946

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0

1

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The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(3)-g(2)}{3-2} = \frac{1-4}{3-2}$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 3 from the left and from the right, $f(x)$ approaches 2 from above. Therefore, the limit of $g(x)$ should be evaluated as x approaches 2 from the right, as shown.

$$\lim_{x \rightarrow 3^-} h(x) = \lim_{x \rightarrow 3^-} g(f(x)) = \lim_{x \rightarrow 2^+} g(x) = 4$$

$$\lim_{x \rightarrow 3^+} h(x) = \lim_{x \rightarrow 3^+} g(f(x)) = \lim_{x \rightarrow 2^+} g(x) = 4$$

Therefore, $\lim_{x \rightarrow 3} h(x) = 4$.

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes two of the criteria below.

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- ☐ continuous at $x = 2$ with justification
- ☐ not continuous at $x = 4$ with justification

Solution:

$$\lim_{x \rightarrow 2^-} k(x) = \lim_{x \rightarrow 2^-} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 2^-} f(x)\right) \lim_{x \rightarrow 2^-} g(x) = (4 - 4)(-1) = 0$$

$$\lim_{x \rightarrow 2^+} k(x) = \lim_{x \rightarrow 2^+} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 2^+} f(x)\right) \lim_{x \rightarrow 2^+} g(x) = (4 - 4)(4) = 0$$

$$k(2) = (4 - f(2))g(2) = (4 - 4)(4) = 0$$

Because $\lim_{x \rightarrow 2} k(x) = 0 = k(2)$, k is continuous at $x = 2$.

$$\lim_{x \rightarrow 4^-} k(x) = \lim_{x \rightarrow 4^-} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 4^-} f(x)\right) \lim_{x \rightarrow 4^-} g(x) = (4 - 3)(1) = 1$$

$$\lim_{x \rightarrow 4^+} k(x) = \lim_{x \rightarrow 4^+} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 4^+} f(x)\right) \lim_{x \rightarrow 4^+} g(x) = (4 - 3)(4) = 4$$

Because $\lim_{x \rightarrow 4^-} k(x) \neq \lim_{x \rightarrow 4^+} k(x)$,

$\lim_{x \rightarrow 4} k(x)$ does not exist and therefore k is not continuous at $x = 4$.

Part D

Any value between 2.784 and 2.787 earns the estimate point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ identifies removable discontinuity
- ☐ estimate
- ☐ reason

Solution:

The graph f of indicates that f has a removable discontinuity at $x = 1$. The graph shows that f is increasing near $x = 1$, so $2.784 < \lim_{x \rightarrow 1} f(x) < 2.787$. A reasonable estimate for $\lim_{x \rightarrow 1} f(x)$ would be determined by averaging the

Unit 1 AP Topics 5-8

function values on each side of 1.

The estimate is as follows.

$$\lim_{x \rightarrow 1} f(x) \approx \frac{f(0.999) + f(1.001)}{2} = \frac{2.784 + 2.787}{2} = 2.7855$$

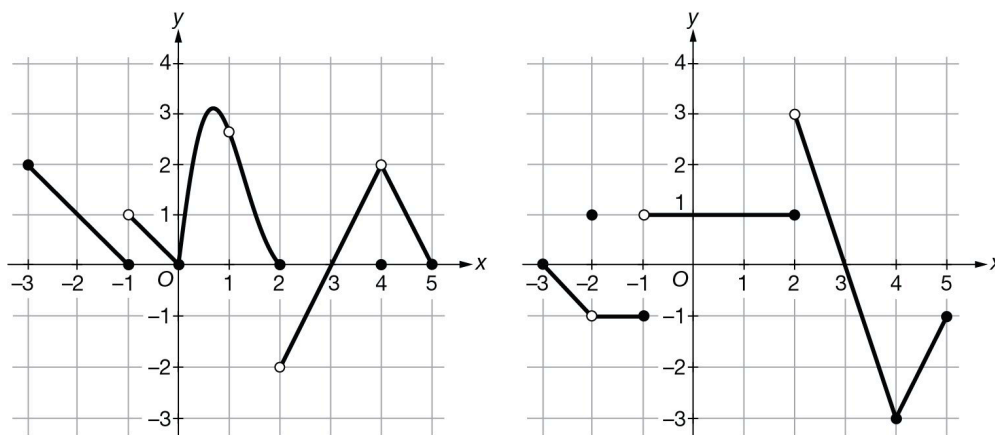
Unit 1 AP Topics 5-8

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f Graph of g

The graphs of the functions f and g are shown above on the interval $-3 \leq x \leq 5$. The graphs consist of line segments, except where $f(x) = \frac{5 \sin(\pi x)}{2 - 2x^3}$ on the interval $0 \leq x \leq 2$.

- (a) Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 4.5$.
- (b) Let $h(x) = g(f(x))$. Find $\lim_{x \rightarrow 4} h(x)$. Use correct limit notation in your answer.
- (c) Let $k(x) = f(x) + g(x)$. Consider $x = -1$, $x = 2$, and $x = 4$. At which one of these values does k have a jump discontinuity? Justify your answer for the jump discontinuity using correct limit notation.
- (d) Values of $f(x)$ at selected values of x are given in the table below. What type of discontinuity does f have at $x = 1$? Based on the values in the table, what is a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$? Give a reason for your answer.

x	0.995	0.996	0.997	0.998	0.999	1	1.001	1.002	1.003	1.004	1.005
$f(x)$	2.631	2.628	2.626	2.623	2.621	Undefined	2.615	2.613	2.610	2.607	2.605

Part A

Select a point value to view scoring criteria, solutions, **and/or** examples to score the response.

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0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(5)-g(4)}{5-4} = \frac{(-1)-(-3)}{5-4}$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 4 from the left and from the right, $f(x)$ approaches 2 from below. Therefore, the limit of $g(x)$ should be evaluated as x approaches 2 from the left, as shown.

$$\lim_{x \rightarrow 4^-} h(x) = \lim_{x \rightarrow 4^-} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) = 1$$

$$\lim_{x \rightarrow 4^+} h(x) = \lim_{x \rightarrow 4^+} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) = 1$$

Therefore, $\lim_{x \rightarrow 4} h(x) = 1$.

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect OR if response includes jump discontinuity at maximum of one incorrect value.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ jump discontinuity at $x = -1$
- ☐ justification, using correct limit notation

Solution:

There is a jump discontinuity at $x = -1$ because both one-sided limits exist at $x = -1$ and are finite but are not equal:

$$\lim_{x \rightarrow -1^-} k(x) = \lim_{x \rightarrow -1^-} f(x) + \lim_{x \rightarrow -1^-} g(x) = 0 + (-1) = -1$$

$$\lim_{x \rightarrow -1^+} k(x) = \lim_{x \rightarrow -1^+} f(x) + \lim_{x \rightarrow -1^+} g(x) = 1 + 1 = 2$$

Note: k does not have a jump discontinuity at $x = 2$. $\lim_{x \rightarrow 2^-} k(x) = 0 + 1 = 1$ and

$$\lim_{x \rightarrow 2^+} k(x) = -2 + 3 = 1$$

Note: k does not have a jump discontinuity at $x = 4$. $\lim_{x \rightarrow 4^-} k(x) = 2 + (-3) = -1$

$$\text{and } \lim_{x \rightarrow 4^+} k(x) = 2 + (-3) = -1$$

Part D

Any value between 2.615 and 2.621 earns the estimate point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ identifies removable discontinuity
- ☐ estimate
- ☐ reason

Unit 1 AP Topics 5-8

Solution:

The graph of f indicates that f has a removable discontinuity at $x = 1$.

The graph shows that f is decreasing near $x = 1$, so $2.621 > \lim_{x \rightarrow 1} f(x) > 2.615$. Since the table of values reflects approximate linear behavior, a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$ would be determined by averaging the function values on each side of 1.

The estimate is as follows. $\lim_{x \rightarrow 1} f(x) \approx \frac{f(0.999) + f(1.001)}{2} = \frac{2.621 + 2.615}{2} = 2.618$

Alternately,

a reasonable estimate can be found by observing that as x increases by 0.001, $f(x)$ decreases by 0.002 or 0.003. Since 2.618 is 0.003 from both 2.621 and 2.615, it is the best estimate.

50.

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1} & \text{for } x \neq 1 \\ 4 & \text{for } x = 1 \end{cases}$$

If f is the function defined as shown, then $\lim_{x \rightarrow 1} f(x)$ is

(A) 0

(B) 2

(C) 4

(D) nonexistent



Answer B

Correct. Looking at the left- and right-sided limits of f at $x = 1$ shows that

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1^-} \frac{(x+1)(x-1)}{x-1} = \lim_{x \rightarrow 1^-} (x+1) = 2 \text{ and}$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1^+} \frac{(x+1)(x-1)}{x-1} = \lim_{x \rightarrow 1^+} (x+1) = 2.$$

Since $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 2$, it follows that $\lim_{x \rightarrow 1} f(x)$ exists and equals 2.

Unit 1 AP Topics 5-8

51. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{x} =$

(A) $-\frac{4}{\pi^2}$

(B) 0

(C) $\frac{2}{\pi}$

(D) 1



Answer C

Correct. Because $\lim_{x \rightarrow \frac{\pi}{2}} x = \frac{\pi}{2} \neq 0$, the limit of the quotient is $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{x} = \frac{\lim_{x \rightarrow \frac{\pi}{2}} \sin x}{\lim_{x \rightarrow \frac{\pi}{2}} x} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$.

52. $\lim_{x \rightarrow 0} \frac{1 - \cos^2(2x)}{(2x)^2} =$

(A) 0

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 1



Answer D

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2(2x)}{(2x)^2} = \lim_{x \rightarrow 0} \frac{\sin^2(2x)}{(2x)^2} = \left(\lim_{x \rightarrow 0} \frac{\sin(2x)}{2x} \right)^2 = \left(\lim_{u \rightarrow 0} \frac{\sin u}{u} \right)^2 = 1^2 = 1$$

Unit 1 AP Topics 5-8

53.

$$f(x) = \begin{cases} \frac{(x-1)^2(x+1)}{|x-1|} & \text{for } x \neq 1 \\ 2 & \text{for } x = 1 \end{cases}$$

If f is the function defined above, then $\lim_{x \rightarrow 1} f(x)$ is

(A) 0

(B) 1

(C) 2

(D) nonexistent



Answer A

Correct. As x approaches 1 from the left, $x - 1 < 0$ and thus $|x - 1| = 1 - x$. Therefore, the left-hand limit can be evaluated as follows: $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{(x-1)^2(x+1)}{|x-1|} = \lim_{x \rightarrow 1^-} \frac{(x-1)^2(x+1)}{1-x} = \lim_{x \rightarrow 1^-} (1-x)(x+1) = 0$.

As x approaches 1 from the right, $x - 1 > 0$ and thus $|x - 1| = x - 1$. Therefore, the right-hand limit can be evaluated as follows: $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{(x-1)^2(x+1)}{|x-1|} = \lim_{x \rightarrow 1^+} \frac{(x-1)^2(x+1)}{x-1} = \lim_{x \rightarrow 1^+} (x-1)(x+1) = 0$.

Since $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 0$, it follows that $\lim_{x \rightarrow 1} f(x) = 0$.

54. It is known that $\lim_{x \rightarrow 0} \frac{\sin(2x)}{2x} = 1$. What is $\lim_{x \rightarrow 0} \frac{\cos(5x)}{8x \cot(2x)}$?

(A) 0

(B) $\frac{1}{8}$ (C) $\frac{1}{4}$

(D) nonexistent



Answer C

Correct. This calculation uses the alternative form of $\cot x$ in terms of sine and cosine.

$$\lim_{x \rightarrow 0} \frac{\cos(5x)}{8x \cot(2x)} = \lim_{x \rightarrow 0} \frac{\cos(5x)}{8x \frac{\cos(2x)}{\sin(2x)}} = \lim_{x \rightarrow 0} \frac{\cos(5x)}{8x} \cdot \frac{\sin(2x)}{\cos(2x)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{4} \cdot \frac{\sin(2x)}{2x} \cdot \frac{\cos(5x)}{\cos(2x)} = \frac{1}{4} \cdot 1 \cdot \frac{1}{1} = \frac{1}{4}$$

Unit 1 AP Topics 5-8

55. $\lim_{x \rightarrow 3} \frac{x-3}{x^3-9x}$ is

(A) 0

(B) $\frac{1}{18}$

(C) 1

(D) nonexistent



Answer B

Correct. To evaluate this limit, it is helpful to first rearrange the expression into an equivalent form by factoring and dividing common factors.

$$\lim_{x \rightarrow 3} \frac{x-3}{x^3-9x} = \lim_{x \rightarrow 3} \frac{x-3}{x(x+3)(x-3)} = \lim_{x \rightarrow 3} \frac{1}{x(x+3)} = \frac{1}{3(6)} = \frac{1}{18}$$

56. If f is the function defined by $f(x) = \frac{x^2-9}{x^2+2x-15}$, then $\lim_{x \rightarrow 3} f(x)$ is

(A) 0

(B) $\frac{9}{15}$

(C) $\frac{3}{4}$

(D) nonexistent



Answer C

Correct. $f(x) = \frac{x^2-9}{x^2+2x-15} = \frac{(x+3)(x-3)}{(x+5)(x-3)} = \frac{x+3}{x+5}$ for x close to but not equal to 3. Therefore,

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x+3}{x+5} = \frac{6}{8} = \frac{3}{4}.$$

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57. If f is the function defined by $f(x) = \frac{x^2-4}{\sqrt{x}-\sqrt{2}}$, then $\lim_{x \rightarrow 2} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow 2} (x+2)(\sqrt{x} + \sqrt{2})$ ✓

(B) $\lim_{x \rightarrow 2} (\sqrt{x} + \sqrt{2})$

(C) $\lim_{x \rightarrow 2} (x\sqrt{x} - 2\sqrt{2})$

(D) $\lim_{x \rightarrow 2} (x+2)(x^2 + 4)$

Answer A

Correct.

$$f(x) = \frac{x^2-4}{\sqrt{x}-\sqrt{2}} = \frac{x^2-4}{\sqrt{x}-\sqrt{2}} \cdot \frac{\sqrt{x}+\sqrt{2}}{\sqrt{x}+\sqrt{2}} = \frac{(x^2-4)(\sqrt{x}+\sqrt{2})}{x-2} = \frac{(x+2)(x-2)(\sqrt{x}+\sqrt{2})}{x-2} = (x+2)(\sqrt{x} + \sqrt{2})$$

for x close to but not equal to 2. Therefore,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x+2)(\sqrt{x} + \sqrt{2}).$$

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58. If g is the function defined by $g(x) = \frac{2\cos^2 x - 1}{\cos x - \sin x}$, then $\lim_{x \rightarrow \frac{\pi}{4}} g(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow \frac{\pi}{4}} (2\cos^2 x - 1)$

(B) $\lim_{x \rightarrow \frac{\pi}{4}} (\cos x + \sin x)$

(C) $\lim_{x \rightarrow \frac{\pi}{4}} (\cos x - \sin x)$

(D) $\frac{\lim_{x \rightarrow \frac{\pi}{4}} (2\cos^2 x - 1)}{\lim_{x \rightarrow \frac{\pi}{4}} (\cos x - \sin x)}$

Answer B

Correct. The numerator and denominator of the expression $\frac{2\cos^2 x - 1}{\cos x - \sin x}$ both approach 0 as x approaches $\frac{\pi}{4}$. The expression can be rewritten using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$, as follows.

$$\frac{2\cos^2 x - 1}{\cos x - \sin x} = \frac{2\cos^2 x - (\cos^2 x + \sin^2 x)}{\cos x - \sin x} = \frac{2\cos^2 x - \cos^2 x - \sin^2 x}{\cos x - \sin x} = \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x} = \frac{(\cos x + \sin x)(\cos x - \sin x)}{\cos x - \sin x} = \cos x + \sin x$$

Therefore, $g(x) = \cos x + \sin x$ for x near $\frac{\pi}{4}$, and

$$\lim_{x \rightarrow \frac{\pi}{4}} g(x) = \lim_{x \rightarrow \frac{\pi}{4}} (\cos x + \sin x).$$

Unit 1 AP Topics 5-8

59.

$$f(x) = \begin{cases} \frac{(x-2)^2 \sin x}{|x-2|} & \text{for } x \neq 2 \\ \pi & \text{for } x = 2 \end{cases}$$

If f is the function defined above, then $\lim_{x \rightarrow 2} f(x)$ is

(A) 0

(B) 1

(C) π

(D) nonexistent



Answer A

Correct. As x approaches 2 from the left, $x - 2 < 0$ and thus $|x - 2| = 2 - x$. Therefore, the left-hand limit can be evaluated as follows:

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{(x-2)^2 \sin x}{|x-2|} = \lim_{x \rightarrow 2^-} \frac{(x-2)(x-2) \sin x}{(2-x)} = \lim_{x \rightarrow 2^-} -(x-2) \sin x = -(2-2) \sin 2 = 0.$$

As x approaches 2 from the right, $x - 2 > 0$ and thus $|x - 2| = x - 2$. Therefore, the right-hand limit can be evaluated as follows:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{(x-2)^2 \sin x}{|x-2|} = \lim_{x \rightarrow 2^+} \frac{(x-2)(x-2) \sin x}{x-2} = \lim_{x \rightarrow 2^+} (x-2) \sin x = (2-2) \sin(2) = 0.$$

Since $\lim_{x \rightarrow 2^-} f(x) = 0 = \lim_{x \rightarrow 2^+} f(x)$, it follows that $\lim_{x \rightarrow 2} f(x) = 0$.

60. $\lim_{x \rightarrow 0} \frac{\cos x + 3e^x}{2e^x}$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) 2

(D) nonexistent



Answer C

Correct. Since $\lim_{x \rightarrow 0} 2e^x \neq 0$, the limit of the quotient is

$$\lim_{x \rightarrow 0} \frac{\cos x + 3e^x}{2e^x} = \frac{\lim_{x \rightarrow 0} (\cos x + 3e^x)}{\lim_{x \rightarrow 0} 2e^x} = \frac{\lim_{x \rightarrow 0} \cos x + 3 \lim_{x \rightarrow 0} e^x}{2 \lim_{x \rightarrow 0} e^x} = \frac{\cos 0 + 3e^0}{2e^0} = \frac{1+3}{2} = 2.$$

Unit 1 AP Topics 5-8

61. If f is the function defined by $f(x) = \frac{x-9}{\sqrt{x}-3}$, then $\lim_{x \rightarrow 9} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow 9} (\sqrt{x} - 3)$

(B) $\lim_{x \rightarrow 9} (\sqrt{x} + 3)$ ✓

(C) $\lim_{x \rightarrow 9} \left(\frac{x^2-81}{x-9} \right)$

(D) $\frac{\lim_{x \rightarrow 9} (x-9)}{\lim_{x \rightarrow 9} (\sqrt{x}-3)}$

Answer B

Correct. $f(x) = \frac{x-9}{\sqrt{x}-3} = \left(\frac{x-9}{\sqrt{x}-3} \right) \left(\frac{\sqrt{x}+3}{\sqrt{x}+3} \right) = \frac{(x-9)(\sqrt{x}+3)}{x-9} = \sqrt{x} + 3$ for x close to, but not equal to, 9.
Therefore, $\lim_{x \rightarrow 9} f(x) = \lim_{x \rightarrow 9} (\sqrt{x} + 3)$.

62. $\lim_{x \rightarrow 0} \frac{7x^5+5x^2+12x}{3x^5+4x}$ is

(A) 0

(B) $\frac{7}{3}$

(C) 3 ✓

(D) ∞

Answer C

Correct. $\lim_{x \rightarrow 0} \frac{7x^5+5x^2+12x}{3x^5+4x} = \lim_{x \rightarrow 0} \frac{x(7x^4+5x+12)}{x(3x^4+4)} = \lim_{x \rightarrow 0} \frac{7x^4+5x+12}{3x^4+4} = \frac{12}{4} = 3$

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63. If $f(x) = \frac{\sin x - 1}{\cos^2 x}$, then $\lim_{x \rightarrow \frac{\pi}{2}} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{-1}{1 + \sin x}$



(B) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{1 + \sin^2 x}$

(C) $\lim_{x \rightarrow \frac{\pi}{2}} \sec x$

(D) $\lim_{x \rightarrow \frac{\pi}{2}} (\tan x - \sec x)$

Answer A

Correct. $f(x) = \frac{\sin x - 1}{\cos^2 x} = \frac{\sin x - 1}{1 - \sin^2 x} = \frac{\sin x - 1}{(1 - \sin x)(1 + \sin x)} = \frac{-1}{1 + \sin x}$ for x close to, but not equal to, $\frac{\pi}{2}$. Therefore $\lim_{x \rightarrow \frac{\pi}{2}} f(x) = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-1}{1 + \sin x}$.

64. If f is the function defined by $f(x) = \frac{\frac{1}{x} - 1}{x - 1}$, then $\lim_{x \rightarrow 1} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow 1} \left(-\frac{1}{x}\right)$



(B) $\lim_{x \rightarrow 1} \left(\frac{1}{x^2} - 1\right)$

(C) $\lim_{x \rightarrow 1} \left(\frac{x-1}{x-1}\right)$

(D) $\frac{\lim_{x \rightarrow 1} (\frac{1}{x} - 1)}{\lim_{x \rightarrow 1} (x - 1)}$

Answer A

Correct. $f(x) = \frac{\frac{1}{x} - 1}{x - 1} = \frac{\frac{1 - x}{x}}{x - 1} = \frac{1 - x}{x(x - 1)} = \frac{-1}{x}$ for x close to, but not equal to, 1. Therefore, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \left(-\frac{1}{x}\right)$.

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65.

$$f(x) = \begin{cases} \frac{x}{|x|} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

If f is the function defined above, then $\lim_{x \rightarrow 0} f(x)$ is

- (A) -1
 (B) 0
 (C) 1
 (D) nonexistent



Answer D

Correct. As x approaches 0 from the left, $x < 0$ and thus $|x| = -x$. Therefore, the left-hand limit can be evaluated as follows: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x}{|x|} = \lim_{x \rightarrow 0^-} \left(\frac{x}{-x}\right) = \lim_{x \rightarrow 0^-} (-1) = -1$.

As x approaches 0 from the right, $x > 0$ and thus $|x| = x$. Therefore, the right-hand limit can be evaluated as follows: $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} \left(\frac{x}{x}\right) = \lim_{x \rightarrow 0^+} 1 = 1$.

Since $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$, it follows that $\lim_{x \rightarrow 0} f(x)$ does not exist.

66. It is known that $\lim_{x \rightarrow 0} \frac{\sin(2x)}{2x} = 1$. What is $\lim_{x \rightarrow 0} \frac{\tan(2x)}{6x \sec(3x)}$?

- (A) 0
 (B) $\frac{1}{6}$
 (C) $\frac{1}{3}$
 (D) nonexistent



Answer C

Correct. This calculation uses the alternative form of $\tan x$ and $\sec x$ in terms of sine and cosine.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan(2x)}{6x \sec(3x)} &= \lim_{x \rightarrow 0} \frac{\frac{\sin(2x)}{\cos(2x)}}{\frac{6x}{\cos(3x)}} = \lim_{x \rightarrow 0} \frac{\sin(2x)}{6x} \cdot \frac{\cos(3x)}{\cos(2x)} \\ &= \lim_{x \rightarrow 0} \frac{1}{3} \cdot \frac{\sin(2x)}{2x} \cdot \frac{\cos(3x)}{\cos(2x)} = \frac{1}{3} \cdot 1 \cdot \frac{1}{1} = \frac{1}{3} \end{aligned}$$

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67. $\lim_{x \rightarrow -4} \frac{x+4}{x^3-16x}$ is

(A) 0

(B) $\frac{1}{32}$

(C) 1

(D) nonexistent

**Answer B**

Correct. To evaluate this limit, it is helpful to first rearrange the expression into an equivalent form by factoring and dividing common factors.

$$\lim_{x \rightarrow -4} \frac{x+4}{x^3-16x} = \lim_{x \rightarrow -4} \frac{x+4}{x(x-4)(x+4)} = \lim_{x \rightarrow -4} \frac{1}{x(x-4)} = \frac{1}{-4(-4-4)} = \frac{1}{32}$$

68. If f is the function defined by $f(x) = \frac{x^2-4}{x^2+x-6}$, then $\lim_{x \rightarrow 2} f(x)$ is

(A) 0

(B) $\frac{2}{3}$

(C) $\frac{4}{5}$

(D) nonexistent

**Answer C**

Correct. $f(x) = \frac{x^2-4}{x^2+x-6} = \frac{(x-2)(x+2)}{(x+3)(x-2)} = \frac{x+2}{x+3}$ for x close to, but not equal to, 2. Therefore,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x+2}{x+3} = \frac{2+2}{2+3} = \frac{4}{5}.$$

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69. If f is the function defined by $f(x) = \frac{x^2-1}{\sqrt{x}-1}$, then $\lim_{x \rightarrow 1} f(x)$ is

(A) 4

(B) 2

(C) 0

(D) nonexistent



Answer A

Correct. $f(x) = \frac{x^2-1}{\sqrt{x}-1} = \frac{x^2-1}{\sqrt{x}-1} \cdot \frac{\sqrt{x}+1}{\sqrt{x}+1} = \frac{(x^2-1)(\sqrt{x}+1)}{x-1} = \frac{(x-1)(x+1)(\sqrt{x}+1)}{x-1} = (x+1)(\sqrt{x}+1)$

for x close to but not equal to 1. Therefore $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x+1)(\sqrt{x}+1) = (1+1)(\sqrt{1}+1) = 4$.

70. If g is the function defined by $g(x) = \frac{\cos x - \sin x}{1 - 2\sin^2 x}$, what is $\lim_{x \rightarrow \frac{\pi}{4}} g(x)$?

(A) 0

(B) $\frac{1}{\sqrt{2}}$

(C) $\sqrt{2}$

(D) The limit does not exist.



Answer B

Correct. The numerator and denominator of the expression $\frac{\cos x - \sin x}{1 - 2\sin^2 x}$ both approach 0 as x approaches $\frac{\pi}{4}$. The expression can be rewritten using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$, as follows.

$$\frac{\cos x - \sin x}{1 - 2\sin^2 x} = \frac{\cos x - \sin x}{\sin^2 x + \cos^2 x - 2\sin^2 x} = \frac{\cos x - \sin x}{\cos^2 x - \sin^2 x} = \frac{\cos x - \sin x}{(\cos x - \sin x)(\cos x + \sin x)} = \frac{1}{\cos x + \sin x}$$

Therefore $g(x) = \frac{1}{\cos x + \sin x}$ for x near $\frac{\pi}{4}$, and $\lim_{x \rightarrow \frac{\pi}{4}} g(x) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{1}{\cos x + \sin x} = \frac{1}{\cos \frac{\pi}{4} + \sin \frac{\pi}{4}} = \frac{1}{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}} = \frac{1}{\sqrt{2}}$.

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71. $\lim_{x \rightarrow 0} \frac{\cos x + 4e^x}{5e^x}$ is

(A) $\frac{1}{5}$

(B) $\frac{4}{5}$

(C) 1

(D) nonexistent



Answer C

Correct. Since $\lim_{x \rightarrow 0} 5e^x \neq 0$, the limit of the quotient is

$$\lim_{x \rightarrow 0} \frac{\cos x + 4e^x}{5e^x} = \frac{\lim_{x \rightarrow 0} (\cos x + 4e^x)}{\lim_{x \rightarrow 0} 5e^x} = \frac{\lim_{x \rightarrow 0} \cos x + \lim_{x \rightarrow 0} 4e^x}{\lim_{x \rightarrow 0} 5e^x} = \frac{\cos 0 + 4 \cdot e^0}{5 \cdot e^0} = \frac{1 + 4 \cdot 1}{5 \cdot 1} = \frac{5}{5} = 1.$$

72. If f is the function defined by $f(x) = \frac{x-4}{\sqrt{x}-2}$, then $\lim_{x \rightarrow 4} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow 4} (\sqrt{x} - 2)$

(B) $\lim_{x \rightarrow 4} (\sqrt{x} + 2)$

(C) $\lim_{x \rightarrow 4} \left(\frac{x^2 - 16}{x - 4} \right)$

(D) $\frac{\lim_{x \rightarrow 4} (x-4)}{\lim_{x \rightarrow 4} (\sqrt{x}-2)}$



Answer B

Correct. $f(x) = \frac{x-4}{\sqrt{x}-2} = \left(\frac{x-4}{\sqrt{x}-2} \right) \left(\frac{\sqrt{x}+2}{\sqrt{x}+2} \right) = \frac{(x-4)(\sqrt{x}+2)}{x-4} = \sqrt{x} + 2$ for x close to, but not equal to, 4.

Therefore, $\lim_{x \rightarrow 4} f(x) = \lim_{x \rightarrow 4} (\sqrt{x} + 2)$.

Alternatively, $\lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x}-2} = \lim_{x \rightarrow 4} \frac{(\sqrt{x}-2)(\sqrt{x}+2)}{\sqrt{x}-2} = \lim_{x \rightarrow 4} (\sqrt{x} + 2)$.

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73. $\lim_{x \rightarrow 0} \frac{5x^5 + 3x^2 + 18x}{3x^5 + 6x}$ is

(A) 0

(B) $\frac{5}{3}$

(C) 3

(D) ∞



Answer C

Correct. $\lim_{x \rightarrow 0} \frac{5x^5 + 3x^2 + 18x}{3x^5 + 6x} = \lim_{x \rightarrow 0} \frac{x(5x^4 + 3x + 18)}{x(3x^4 + 6)} = \lim_{x \rightarrow 0} \frac{5x^4 + 3x + 18}{3x^4 + 6} = \frac{0 + 0 + 18}{0 + 6} = 3$

74. If $f(x) = \frac{\cos x - 1}{\sin^2 x}$, then $\lim_{x \rightarrow 0} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow 0} \frac{-1}{1 + \cos x}$

(B) $\lim_{x \rightarrow 0} \frac{\cos x - 1}{1 + \cos^2 x}$

(C) $\lim_{x \rightarrow 0} \csc x$

(D) $\lim_{x \rightarrow 0} (\cot x - \csc x)$



Answer A

Correct. $f(x) = \frac{\cos x - 1}{\sin^2 x} = \frac{\cos x - 1}{1 - \cos^2 x} = \frac{\cos x - 1}{(1 + \cos x)(1 - \cos x)} = \frac{-1}{1 + \cos x}$ for x close to, but not equal to, 0. Therefore,
 $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{-1}{1 + \cos x}.$

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75.

$$f(x) = \begin{cases} \frac{|x|}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

If f is the function defined above, then $\lim_{x \rightarrow 0} f(x)$ is

- (A) -1
(B) 0
(C) 1
(D) nonexistent

**Answer D**

Correct. By definition

$$|x| = \begin{cases} -x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$

As x approaches 0 from the left, $x < 0$ and thus $|x| = -x$. Therefore, the left-hand limit can be evaluated as follows: $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \left(\frac{-x}{x} \right) = \lim_{x \rightarrow 0^-} (-1) = -1$.

As x approaches 0 from the right, $x > 0$ and thus $|x| = x$. Therefore, the right-hand limit can be evaluated as follows: $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \left(\frac{x}{x} \right) = \lim_{x \rightarrow 0^+} 1 = 1$.

Since $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$, it follows that $\lim_{x \rightarrow 0} f(x)$ does not exist.

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76. $\lim_{x \rightarrow 4} \frac{2 - \sqrt{x}}{x - 4}$ is

(A) $-\frac{1}{4}$

(B) 0

(C) $\frac{1}{4}$

(D) nonexistent

Answer A

Correct. The given expression is multiplied by $\frac{2+\sqrt{x}}{2+\sqrt{x}}$ to rearrange the given expression into an equivalent form:

$$\frac{2-\sqrt{x}}{x-4} = \frac{2-\sqrt{x}}{x-4} \cdot \frac{2+\sqrt{x}}{2+\sqrt{x}} = \frac{4-x}{(x-4)(2+\sqrt{x})} = \frac{-1}{2+\sqrt{x}}. \text{ It follows that } \lim_{x \rightarrow 4} \frac{-1}{2+\sqrt{x}} = -\frac{1}{4}.$$

Alternatively, after the minus sign has been factored out in the numerator, the limit can be written in the form of the difference quotient $\lim_{x \rightarrow 4} \frac{f(x)-f(4)}{x-4}$, where $f(x) = \sqrt{x}$. This limit is one way to express the derivative of f at $x = 4$.

$$\text{Since } f'(x) = \frac{1}{2\sqrt{x}}, \lim_{x \rightarrow 4} \frac{f(4)-f(x)}{x-4} = -\left(\lim_{x \rightarrow 4} \frac{f(x)-f(4)}{x-4}\right) = -f'(4) = -\frac{1}{2\sqrt{4}} = -\frac{1}{4}.$$

77. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6}$ is

(A) -4

(B) 0

(C) 4

(D) nonexistent

Answer A

Correct. The limit can be found using algebraic manipulation, as follows.

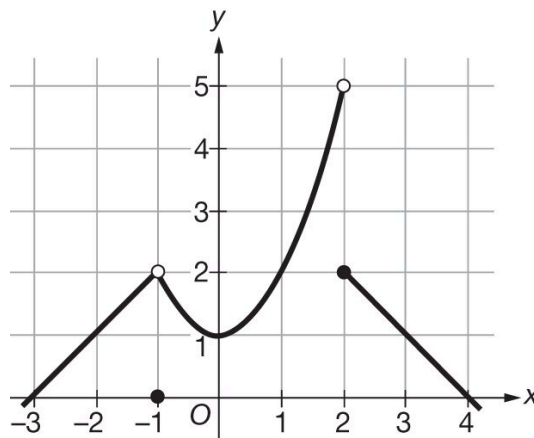
$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)(x-3)} = \lim_{x \rightarrow 2} \frac{x+2}{x-3} = \frac{4}{-1} = -4$$

Alternatively, since $\lim_{x \rightarrow 2} (x^2 - 4) = 0$ and $\lim_{x \rightarrow 2} (x^2 - 5x + 6) = 0$, L'Hospital's rule can be used with this indeterminate form, as follows.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6} = \lim_{x \rightarrow 2} \frac{2x}{2x - 5} = \frac{4}{-1} = -4$$

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78.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow -1} f(f(x))$?

- (A) 1
(B) 2
(C) 5
(D) nonexistent

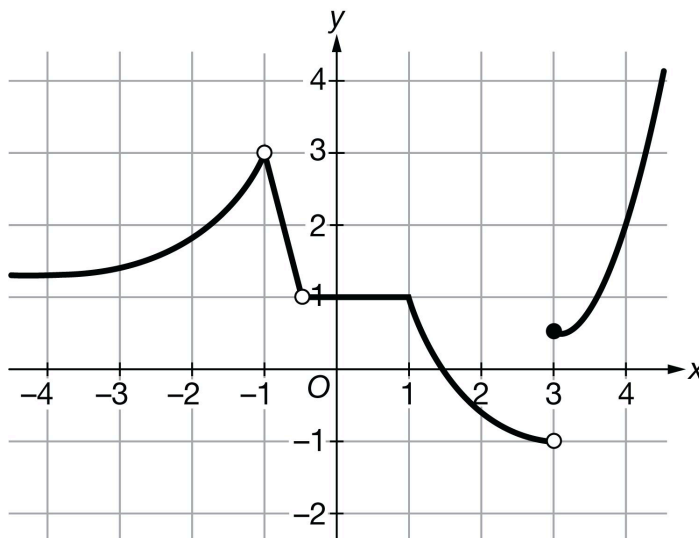
**Answer C**

Correct. The important observation is that as x approaches -1 , $f(x)$ approaches 2 in the graph from below. Therefore, the limit of $f(f(x))$ should be evaluated as x approaches 2 from the left, as shown.

$$\lim_{x \rightarrow -1} f(f(x)) = \lim_{x \rightarrow 2^-} f(x) = 5.$$

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79.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow -1} f(f(x))$?

- (A) 3
(B) $\frac{1}{2}$
(C) -1
(D) nonexistent

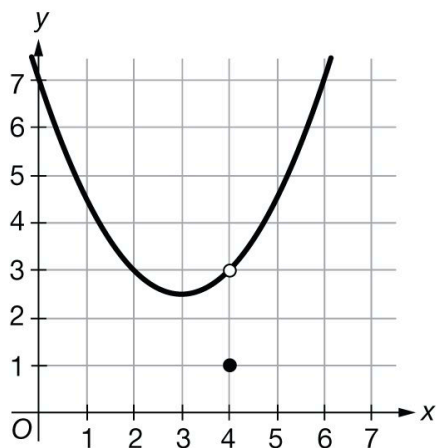
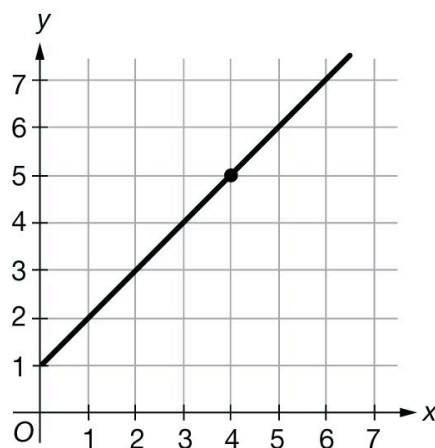
**Answer C**

Correct. The important observation is that as x approaches -1 , $f(x)$ approaches 3 in the graph from the left. Therefore, the limit of $f(x)$ should be evaluated as x approaches 3 from the left, as shown.

$$\lim_{x \rightarrow -1} f(f(x)) = \lim_{x \rightarrow 3^-} f(x) = -1$$

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80.

Graph of f Graph of g

The graphs of the functions f and g are shown above. The value of $\lim_{x \rightarrow 4} \frac{f(x)+7}{g(x)}$ is

(A) $\frac{3}{5}$

(B) $\frac{8}{5}$

(C) 2

(D) nonexistent

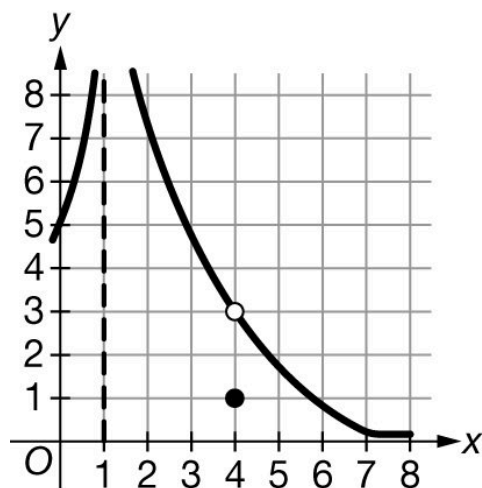
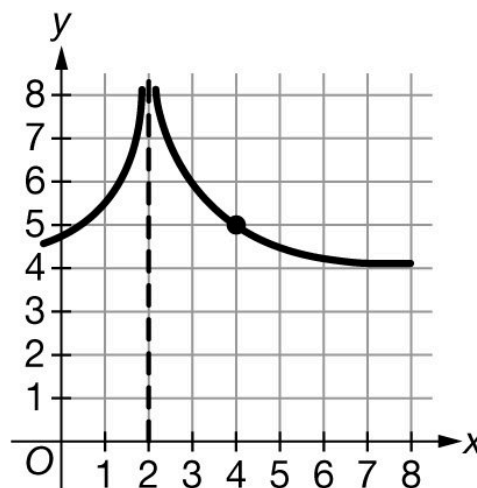
**Answer C**

Correct. Since $\lim_{x \rightarrow 4} f(x) = 3$ and $\lim_{x \rightarrow 4} g(x) = 5 \neq 0$, the limit of the quotient is

$$\lim_{x \rightarrow 4} \frac{f(x)+7}{g(x)} = \frac{\left(\lim_{x \rightarrow 4} f(x)\right)+7}{\lim_{x \rightarrow 4} g(x)} = \frac{3+7}{5} = 2.$$

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81.

Graph of f Graph of g

The graphs of the functions f and g are shown above. The value of $\lim_{x \rightarrow 4} \frac{f(x)+6}{g(x)}$ is

(A) $\frac{3}{5}$

(B) $\frac{7}{5}$

(C) $\frac{9}{5}$

(D) nonexistent

**Answer C**

Correct. Since $\lim_{x \rightarrow 4} f(x) = 3$ and $\lim_{x \rightarrow 4} g(x) = 5 \neq 0$, the limit of the quotient is

$$\lim_{x \rightarrow 4} \frac{f(x)+6}{g(x)} = \frac{\left(\lim_{x \rightarrow 4} f(x)\right)+6}{\lim_{x \rightarrow 4} g(x)} = \frac{3+6}{5} = \frac{9}{5}.$$

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82. Let f be the function defined by $f(x) = \frac{(x-2)(x^2-k^2)}{(x^2-4)(x-k)}$, where k is a constant. For what value of k , if any, does $\lim_{x \rightarrow 2} f(x) = 3$?
- (A) 2
(B) 3
(C) 10
(D) There is no such value of k .

Answer C

Correct. The limit can be found using algebraic manipulation to rearrange the expression into an equivalent form, as follows.

$$\begin{aligned}\lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2-k^2)}{(x^2-4)(x-k)} = \lim_{x \rightarrow 2} \frac{(x-2)(x-k)(x+k)}{(x-2)(x+2)(x-k)} = \lim_{x \rightarrow 2} \frac{x+k}{x+2} = \frac{2+k}{4} \\ \lim_{x \rightarrow 2} f(x) &= 3 \Rightarrow \frac{2+k}{4} = 3 \Rightarrow k = 10\end{aligned}$$

83.
$$f(x) = \begin{cases} 5x - 3 & \text{for } x < 2 \\ 9 & \text{for } x = 2 \\ 4x + 3 & \text{for } x > 2 \end{cases}$$

Let f be the piecewise function defined above. The value of $\lim_{x \rightarrow 2^+} f(x)$ is

- (A) 7
(B) 9
(C) 11
(D) nonexistent

Answer C

Correct. $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (4x + 3) = 4 \cdot 2 + 3 = 11$

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84.

$$f(x) = \begin{cases} 2x + 2 & \text{for } x < 1 \\ x^2 + 4 & \text{for } x > 1 \end{cases}$$

If f is the function defined above, then $\lim_{x \rightarrow 1^-} f(x)$ is

(A) 2

(B) 4

(C) 5

(D) nonexistent

**Answer B**

Correct. $\lim_{x \rightarrow 1^-} f(x)$ is the limit of $f(x)$ as x approaches 1 from the left. $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 2x + 2 = 2 + 2 = 4$

85.

$$f(x) = \begin{cases} x^2 - 4x + 4 & \text{for } x < 2 \\ 3 & \text{for } x = 2 \\ x^2 - 4x + 8 & \text{for } x > 2 \end{cases}$$

Let f be the piecewise function defined above. The value of $\lim_{x \rightarrow 2^+} f(x)$ is

(A) 0

(B) 3

(C) 4

(D) nonexistent

**Answer C**

Correct. $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} x^2 - 4x + 8 = 4 - 4 \cdot 2 + 8 = 4$

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86.

$$f(x) = \begin{cases} x + 3 & \text{for } x < 1 \\ -2x + 7 & \text{for } x > 1 \end{cases}$$

If f is the function defined above, then $\lim_{x \rightarrow 1^-} f(x)$ is

(A) 2

(B) 4

(C) 5

(D) nonexistent

**Answer B**

Correct. $\lim_{x \rightarrow 1^-} f(x)$ is the limit of $f(x)$ as x approaches 1 from the left.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x + 3) = 1 + 3 = 4$$

87.

$f(2) = 3$	$\lim_{x \rightarrow 2} f(x) = 4$
$g(2) = -6$	$\lim_{x \rightarrow 2} g(x) = -6$
$h(2) = -3$	$\lim_{x \rightarrow 2} h(x) = 2$

The table above gives selected values and limits of the functions f , g , and h . What is $\lim_{x \rightarrow 2} (h(x)(5f(x) + g(x)))$?

(A) -27

(B) -20

(C) 28

(D) 34

**Answer C**

Correct. $\lim_{x \rightarrow 2} (h(x)(5f(x) + g(x))) = \lim_{x \rightarrow 2} h(x) \cdot \left(5 \cdot \lim_{x \rightarrow 2} f(x) + \lim_{x \rightarrow 2} g(x) \right) = 2 \cdot (5 \cdot 4 - 6) = 28$

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88. Let f and g be functions such that $\lim_{x \rightarrow 4} g(x) = 2$ and $\lim_{x \rightarrow 4} \frac{f(x)}{g(x)} = \pi$. What is $\lim_{x \rightarrow 4} f(x)$?

(A) $\frac{\pi}{2}$

(B) $2 + \pi$

(C) 2π

(D) The limit cannot be determined from the information given.

Answer C

Correct. Because $\lim_{x \rightarrow 4} g(x) \neq 0$, the quotient property for limits can be used, as follows:

$$\pi = \lim_{x \rightarrow 4} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow 4} f(x)}{\lim_{x \rightarrow 4} g(x)} = \frac{\lim_{x \rightarrow 4} f(x)}{2}. \text{ Therefore, } \lim_{x \rightarrow 4} f(x) = 2\pi.$$

89. If f is the function defined by $f(x) = \frac{x-1}{1-\frac{1}{x}}$, then $\lim_{x \rightarrow 1} f(x)$ is equivalent to which of the following?

(A) $\lim_{x \rightarrow 1} x$

(B) $\lim_{x \rightarrow 1} 2x$

(C) $\lim_{x \rightarrow 1} \left(\frac{x-1}{1-x} \right)$

(D) $\frac{\lim_{x \rightarrow 1} (x-1)}{\lim_{x \rightarrow 1} \left(1 - \frac{1}{x} \right)}$

Answer A

Correct. $f(x) = \frac{x-1}{1-\frac{1}{x}} = \frac{x-1}{\frac{x-1}{x}} = \frac{x-1}{x} \cdot \frac{x}{x-1} = x$ for x close to, but not equal to, 1. Therefore, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} x$.

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90. Let f and g be functions such that $\lim_{x \rightarrow 4} g(x) = 7$ and $\lim_{x \rightarrow 4} \frac{f(x)}{g(x)} = \pi$. What is $\lim_{x \rightarrow 4} f(x)$?

(A) $\frac{\pi}{7}$

(B) $7 + \pi$

(C) 7π

(D) The limit cannot be determined from the information given.



Answer C

Correct. Because $\lim_{x \rightarrow 4} g(x) \neq 0$, the quotient property for limits can be used, as follows:

$$\pi = \lim_{x \rightarrow 4} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow 4} f(x)}{\lim_{x \rightarrow 4} g(x)} = \frac{\lim_{x \rightarrow 4} f(x)}{7}. \text{ Therefore, } \lim_{x \rightarrow 4} f(x) = 7\pi.$$

91.

$f(3) = 3$	$\lim_{x \rightarrow 3} f(x) = 2$
$g(3) = 8$	$\lim_{x \rightarrow 3} g(x) = 8$
$h(3) = 4$	$\lim_{x \rightarrow 3} h(x) = 2$

The table above gives selected values and limits of the functions f , g , and h . What is $\lim_{x \rightarrow 3} (h(x)(2f(x) + 3g(x)))$?

(A) 120

(B) 104

(C) 56

(D) 32



Answer C

Correct.

$$\lim_{x \rightarrow 3} (h(x)(2f(x) + 3g(x))) = \lim_{x \rightarrow 3} h(x) \cdot \left(2 \cdot \lim_{x \rightarrow 3} f(x) + 3 \cdot \lim_{x \rightarrow 3} g(x) \right) = 2 \cdot (2 \cdot 2 + 3 \cdot 8) = 2(4 + 24) = 56$$

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92. Let f and g be the functions defined by $f(x) = \frac{x^2-25}{x^2-29}$ and $g(x) = \frac{x+5}{2x+4}$. If the function h satisfies $f(x) \leq h(x) \leq g(x)$ for $1 \leq x \leq 5$, what is $\lim_{x \rightarrow 3} h(x)$?

(A) $\frac{1}{2}$

(B) $\frac{4}{5}$

(C) 1

(D) The limit cannot be determined from the given information.

Answer B

Correct. Because $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x^2-25}{x^2-29} = \frac{9-25}{9-29} = \frac{-16}{-20} = \frac{4}{5}$, $\lim_{x \rightarrow 3} g(x) = \lim_{x \rightarrow 3} \frac{x+5}{2x+4} = \frac{3+5}{6+4} = \frac{8}{10} = \frac{4}{5}$, and $f(x) \leq h(x) \leq g(x)$ on the interval $1 \leq x \leq 5$, the squeeze theorem implies that $\lim_{x \rightarrow 3} h(x) = \frac{4}{5}$.